

#1 · Common Occupational Skills

What major hazard categories does WHMIS 2015 recognize?

Answer

WHMIS 2015 adopts the Globally Harmonized System (GHS) and groups hazards into three broad categories: physical hazards (such as flammable gases, oxidizing gases, compressed gas, flammable liquids, flammable solids, and self-reactive materials), health hazards (such as acute toxicity, skin corrosion, eye damage, respiratory sensitization, carcinogenicity, and reproductive toxicity), and environmental hazards. There are more than nine individual classes in total; each class has a distinct GHS pictogram.

#2 · Common Occupational Skills

Which SDS section lists first aid measures for a chemical?

Answer

Section 4 of a Safety Data Sheet covers first aid measures. It tells you what to do if someone swallows, inhales, or gets a chemical on their skin or eyes. Always read this section before you start working with a new product.

#3 · Common Occupational Skills

What does SDS Section 8 tell you about a chemical?

Answer

Section 8 covers exposure controls and personal protective equipment. It lists occupational exposure limits (OELs) and recommends the specific gloves, respirator, eye protection, and clothing required. Use this section to select the right PPE before starting work.

#4 · Common Occupational Skills

When do you choose a P100 respirator filter over an N95?

Answer

A P100 filter is oil-proof and blocks 99.97% of airborne particles, making it the right choice when oil mist or fine metallic dust is present. An N95 is not oil-resistant and only blocks 95% of non-oil particles. Match the filter class to the hazard in the SDS.

#5 · Common Occupational Skills

What PPE is required when grinding metal in a shop?

Answer

You need safety glasses plus a full face shield to protect against flying sparks and fragments. Leather gloves protect your hands from sharp edges and heat. A hearing protector is also needed because grinding noise often exceeds 85 dB.

#6 · Common Occupational Skills

List the six steps of Lockout/Tagout (LOTO) in order.

Answer

The six steps are: identify all energy sources, notify affected workers, shut down the equipment, isolate every energy source with locks, release or restrain stored energy (bleed lines, block gravity loads, discharge capacitors), and verify zero energy state before touching the machine. Each worker applies their own lock.

#7 · Common Occupational Skills

What is stored energy and why must it be released during LOTO?

Answer

Stored energy is energy trapped in the machine after shutdown, such as compressed springs, pressurized hydraulic lines, elevated loads, or charged capacitors. It can still move parts and injure workers even when the power is off. Releasing or blocking stored energy is a mandatory LOTO step.

#8 · Common Occupational Skills

What are the three main hazards in a permit-required confined space?

Answer

The three main hazards are: atmospheric (toxic gas, oxygen deficiency, or flammable vapour), engulfment (flowing solids or liquids that can trap a person), and configuration (internal shape that can trap or asphyxiate). All three must be assessed and controlled before entry. A rescue plan and attendant outside are always required.

#9 · Common Occupational Skills

What oxygen level is considered safe for confined space entry?

Answer

A safe oxygen level is between 19.5% and 23.0% by volume. Below 19.5% is oxygen-deficient and can cause unconsciousness quickly. Above 23.0% is oxygen-enriched and greatly increases fire and explosion risk.

#10 · Common Occupational Skills

What information must a hot work permit include before work begins?

Answer

A hot work permit must include the exact work location, the type of hot work (welding, cutting, grinding), the date and time window, fire watch requirements, combustibles removed or covered, and the name of the person who authorized the permit. It must be posted at the work site.

#11 · Common Occupational Skills

What does SWL mean and how does it differ from WLL?

Answer

SWL (Safe Working Load) is an older term for the maximum load a piece of rigging hardware should carry under normal use. WLL (Working Load Limit) is the modern replacement term with the same meaning. Both are stamped or tagged on the hardware by the manufacturer and must never be exceeded.

#12 · Common Occupational Skills

How do you convert a fraction like $\frac{7}{32}$ to a decimal?

Answer

Divide the top number (numerator) by the bottom number (denominator): 7 divided by 32 equals 0.21875. This is useful when you need to enter a fractional measurement into a calculator or compare it to a metric value. Practice this conversion for common drill and shim sizes.

#13 · Common Occupational Skills

What is the formula to convert inches to millimetres?

Answer

Multiply the inch value by 25.4 to get millimetres. For example, 2.5 inches times 25.4 equals 63.5 mm. To go the other way, divide millimetres by 25.4 to get inches.

#14 · Common Occupational Skills

A shaft is dimensioned 50.000 mm with a tolerance of plus 0.025, minus 0.000. What are the limits?

Answer

The maximum shaft size is 50.025 mm and the minimum is 50.000 mm. The total tolerance band is 0.025 mm. Any shaft outside this range does not meet the drawing specification and must be rejected or reworked.

#15 · Common Occupational Skills

What does the blueprint symbol for a hidden line look like?

Answer

A hidden line is drawn as a series of short dashes, evenly spaced, usually about 3 mm long with 1.5 mm gaps. It shows an edge or feature that exists behind a surface you cannot see from that view. Hidden lines help you understand the full 3D shape without needing a cutaway view.

#16 · Common Occupational Skills

What does a centre line symbol on a blueprint represent?

Answer

A centre line is drawn as alternating long and short dashes. It marks the exact centre of a hole, shaft, or symmetrical feature. Dimensions to holes are almost always measured from centre lines, so reading them correctly is critical for accurate layout.

#17 · Common Occupational Skills

What does the GD&T; flatness symbol control on a part?

Answer

Flatness controls how flat a single surface is, with no reference to any datum. The tolerance zone is the space between two perfectly flat parallel planes. The entire surface must fit inside that zone, regardless of where the part sits in space.

#18 · Common Occupational Skills

What does GD&T; roundness (circularity) control?

Answer

Roundness controls how close to a perfect circle any cross-section of a cylindrical feature is. The tolerance zone is the annular space between two concentric circles. It is checked at individual cross-sections, not along the full length of the feature.

#19 · Common Occupational Skills

What does GD&T; perpendicularity control and what does it require?

Answer

Perpendicularity controls how close to a perfect 90-degree angle a surface or axis is relative to a datum. It always requires a datum reference. The tolerance zone can be two parallel planes or a cylinder, depending on whether a surface or an axis is being controlled.

#20 · Common Occupational Skills

What does GD&T; parallelism control on a drawing?

Answer

Parallelism controls how close a surface or axis is to being perfectly parallel to a datum. Like perpendicularity, it always needs a datum. The tolerance zone is two planes parallel to the datum, and the entire controlled feature must fall within that zone.

#21 · Common Occupational Skills

What does a surface finish symbol with a value of Ra 1.6 mean?

Answer

Ra is the average roughness of a machined surface, measured in micrometres. Ra 1.6 means the average peak-to-valley height is 1.6 micrometres, which is a smooth finish typical of a ground surface. Lower Ra numbers mean smoother surfaces; higher numbers mean rougher ones.

#22 · Common Occupational Skills

What is the difference between an interference fit and a clearance fit?

Answer

In an interference fit, the shaft is always larger than the hole, so the parts must be pressed or heated to assemble. This creates a tight, permanent connection with no play. In a clearance fit, the shaft is always smaller than the hole, so parts slide together easily and can be disassembled without force.

#23 · Common Occupational Skills

What is tolerance stackup and why does it matter to a millwright?

Answer

Tolerance stackup happens when individual part tolerances add up across an assembly, potentially pushing the total variation beyond acceptable limits. Even if every part is within its own tolerance, the combined error can cause misalignment or binding. Millwrights check stackup when fitting multiple shims or spacers in series.

#24 · Common Occupational Skills

What is the formula for calculating torque from force and lever arm?

Answer

Torque equals force multiplied by the perpendicular distance from the pivot point to the line of force ($T = F \times d$). Force is measured in Newtons and distance in metres, giving torque in Newton-metres (Nm). A longer wrench handle produces more torque with the same hand force.

#25 · Common Occupational Skills

How do you read a 0-25 mm outside micrometer to 0.001 mm?

Answer

Read the sleeve scale for the whole and half millimetres (each line is 0.5 mm). Read the thimble scale for hundredths (each graduation is 0.01 mm). Then read the vernier line on the sleeve that aligns with a thimble line for thousandths (each vernier division is 0.001 mm). Add all three readings together.

#26 · Common Occupational Skills

What is the measuring range of a standard vernier caliper and its resolution?

Answer

A standard vernier caliper measures from 0 to 150 mm (or 0 to 6 inches) and resolves to 0.02 mm with a 50-division vernier scale. It measures outside diameter, inside diameter, depth, and step height using its four measuring faces. The digital version reads to 0.01 mm.

#27 · Common Occupational Skills

What does a dial indicator measure and what is its typical resolution?

Answer

A dial indicator measures small amounts of movement or deviation at a point, typically resolving to 0.01 mm (0.001 inch on imperial versions). Millwrights use them to check shaft runout, bearing clearance, and machine alignment. The indicator must be mounted rigidly; any flex in the stand gives a false reading.

#28 · Common Occupational Skills

What is a telescoping gauge and how do you use it?

Answer

A telescoping gauge is a two-armed spring-loaded tool that expands to contact the walls of a bore or slot. You compress it, insert it into the hole, lock it, and withdraw it, then measure across the arms with a micrometer. It transfers the bore size to the micrometer for an accurate reading.

#29 · Common Occupational Skills

What is a bore gauge and when do you use it instead of a telescoping gauge?

Answer

A bore gauge (also called an internal dial indicator or three-point bore gauge) measures the diameter of a bore directly in one step without a transfer measurement. It is more accurate and faster for repeated measurements in production settings. Use it when bore size must be held to a tight tolerance, such as when fitting a bearing.

#30 · Common Occupational Skills

What is a datum reference in GD&T; and why is it used?

Answer

A datum is a theoretically perfect point, line, axis, or plane used as the starting point for measurements. Physical datum features on the part (like a machined surface or bore) simulate this perfect reference. All location and orientation tolerances are measured from datums so that every inspector gets the same result.

#31 · Common Occupational Skills

What does chain of custody mean for a calibrated measuring tool?

Answer

Chain of custody means tracking every person who uses, stores, or services a calibrated tool from its last calibration to its next one. If the chain is broken (tool dropped, lost, or used by an unauthorized person), the tool must be recalibrated before use. This ensures measurement traceability and prevents bad data from entering production.

#32 · Common Occupational Skills

How often should a micrometer be calibrated and what is the standard?

Answer

Calibration frequency depends on the shop's quality system, but a common standard is every 6 to 12 months or after any impact or suspected damage. Calibration is done against gauge blocks traceable to national standards (such as NRC in Canada or NIST in the US). A calibration sticker on the tool shows the next due date.

#33 · Common Occupational Skills

What is the purpose of a feeler gauge in millwright work?

Answer

A feeler gauge is a set of thin metal blades of known thickness used to measure small gaps, such as bearing clearance, gear backlash, or valve clearance. You select the blade (or combination of blades) that fits snugly in the gap with a slight drag. The blade thickness equals the gap size.

#34 · Common Occupational Skills

What is a height gauge used for on a surface plate?

Answer

A height gauge measures vertical distances from a reference surface plate to a feature on a part, or scribes layout lines at a precise height. The base slides on the surface plate and the measuring arm or scribe is set to the required height. Surface plates must be clean and undamaged for accurate measurements.

#35 · Common Occupational Skills

What is an angle gauge block and when would a millwright use one?

Answer

An angle gauge block is a precision-ground block made to an exact angle, used to set or check angular features on machine tools or fixtures. Millwrights use them to verify the taper of a shaft, set a compound slide, or check a vee-block. They are wrung together like linear gauge blocks to build compound angles.

#36 · Rigging Hoisting and Moving

What does 6x19 wire rope construction mean?

Answer

6x19 means the rope has 6 strands, and each strand is made from approximately 19 individual wires. This construction balances flexibility and strength, making it common for general rigging and crane hoist lines. More wires per strand (like 6x37) gives more flexibility but less resistance to abrasion.

#37 · Rigging Hoisting and Moving

What is 6x37 wire rope typically used for compared to 6x19?

Answer

6x37 wire rope has 6 strands of about 37 wires each, making it much more flexible than 6x19. It is used on cranes with small sheave diameters or where the rope must bend sharply and repeatedly. The finer wires wear faster on rough surfaces, so it is less suited for dragging or abrasive conditions.

#38 · Rigging Hoisting and Moving

How many broken wires in one lay length require wire rope retirement?

Answer

If there are 6 or more broken wires in one rope lay length in any one strand, or 12 or more broken wires distributed across all strands in one lay length, the rope must be taken out of service. A rope lay length is approximately the length of one full helical turn of a strand. These are standard criteria from CSA and ASME B30.

#39 · Rigging Hoisting and Moving

What is birdcaging in wire rope and what causes it?

Answer

Birdcaging is a defect where the outer strands of a wire rope splay outward in a cage shape, resembling a birdcage. It is caused by a sudden release of a heavy load or shock loading, which unwinds the strands. A birdcaged rope must be removed from service immediately because it has lost its rated strength.

#40 · Rigging Hoisting and Moving

What is the sling angle factor and why does it reduce the WLL?

Answer

When a sling is not vertical, the tension in each leg is higher than the weight being lifted because the load force is shared at an angle. The sling angle factor accounts for this: at 60 degrees from horizontal, each sling leg carries 115% of what it would carry if vertical. At 30 degrees, each leg carries 200%, which can easily exceed the WLL.

#41 · Rigging Hoisting and Moving

What is the minimum acceptable sling angle from horizontal during a lift?

Answer

The minimum recommended sling angle is 45 degrees from horizontal. Below 45 degrees, the tension in the sling legs rises sharply and can exceed the WLL even with a moderate load. The ideal angle is 60 degrees or greater to keep tension within safe limits.

#42 · Rigging Hoisting and Moving

What is the difference between a vertical hitch, choker hitch, and basket hitch?

Answer

A vertical hitch uses a single sling with one end attached to the crane hook and the other to the load, providing 100% of the sling's WLL. A choker hitch wraps around the load and through its own eye, giving about 75-80% WLL due to the bend angle. A basket hitch passes under the load with both ends on the hook, giving up to 200% WLL when the legs are vertical.

#43 · Rigging Hoisting and Moving

What is the difference between a screw-pin shackle and a bolt-type shackle?

Answer

A screw-pin shackle uses a threaded pin that you hand-tighten and is quick to rig and de-rig, but the pin can back out under repeated side loading or vibration. A bolt-type (safety) shackle uses a bolt with a nut and cotter pin, which locks the pin in place. Use bolt-type shackles in permanent or semi-permanent rigging where vibration is present.

#44 · Rigging Hoisting and Moving

What does proof testing rigging hardware mean?

Answer

Proof testing means applying a test load (typically twice the WLL) to a piece of rigging hardware after manufacture or repair to verify it can hold that load without permanent deformation or failure. Hardware that passes proof testing is stamped with the WLL. Field riggers should never proof test hardware themselves; it is done by the manufacturer.

#45 · Rigging Hoisting and Moving

What crane hand signal tells the operator to raise the boom?

Answer

To signal raise the boom, the rigger extends one arm horizontally with the thumb pointing upward. All crane hand signals must be agreed upon and reviewed before the lift. Only one designated signal person gives signals to the operator at any time.

#46 · Rigging Hoisting and Moving

What crane hand signal tells the operator to stop all movement immediately?

Answer

The emergency stop signal is both arms extended horizontally at shoulder height with palms facing down, moved rapidly back and forth. Any worker on site can give this signal, not just the designated signal person. The operator must stop immediately when they see it.

#47 · Rigging Hoisting and Moving

What makes a lift a critical lift requiring a written lift plan?

Answer

A critical lift is generally defined as a lift that exceeds 75% of the crane's rated capacity, involves multiple cranes working together, requires lifting over energized equipment or personnel, involves unusual load shapes, or has very limited crane positioning. The exact threshold varies by jurisdiction but a written lift plan, engineer involvement, and a pre-lift meeting are required.

#48 · Rigging Hoisting and Moving

How do you read a crane load chart to find the allowable lift capacity?

Answer

First find the crane's current configuration (boom length, counterweight, outrigger setting). Then locate the radius (horizontal distance from the crane centre pin to the load) on the chart's left column. Read across to the column matching the boom length to find the allowable load in that configuration. Always use the most restrictive value if the chart has multiple configurations.

#49 · Rigging Hoisting and Moving

What is the purpose of a tag line during a crane lift?

Answer

A tag line is a rope attached to the load and held by a ground worker to control the load's rotation and drift as it is lifted and moved. It prevents the load from spinning into structures or workers. Tag lines should never be wrapped around a worker's hand or body.

#50 · Rigging Hoisting and Moving

What is a spreader bar used for in rigging?

Answer

A spreader bar holds the sling attachment points apart, increasing the sling angle and reducing compressive force on a long or fragile load. Without a spreader bar, slings on a long load angle inward and can crush the load or slip off. The bar itself must be rated for the compressive load it receives.

#51 · Rigging Hoisting and Moving

How does the load centre of gravity affect forklift capacity?

Answer

Forklift capacity is rated at a standard load centre (typically 600 mm or 24 inches from the fork face). If the load's centre of gravity is farther from the forks, the effective capacity drops because the load creates a larger tipping moment. Always check the forklift's load capacity chart for the actual load centre you are working with.

#52 · Rigging Hoisting and Moving

What is the difference between nylon and polyester synthetic slings?

Answer

Nylon slings stretch under load (about 6-10% elongation), which can be useful for shock-absorbing lifts but causes the load to drop slightly during pick-up. Polyester slings stretch very little (1-3%), making them better when precise load control is needed. Polyester also resists acid and bleach better than nylon, while nylon handles alkalis better.

#53 · Rigging Hoisting and Moving

When must a synthetic web sling be removed from service?

Answer

Remove a synthetic sling if it has cuts, holes, tears, abrasion that exposes the load-bearing yarns, acid or heat damage (discolouration, brittleness), knots, or a missing or illegible tag. You cannot repair synthetic slings in the field. When in doubt, take it out.

#54 · Rigging Hoisting and Moving

How do you find the centre of gravity of an irregular load before a lift?

Answer

For an irregular load, balance the load on a single point (like a pipe or knife-edge block) along each axis and mark where it balances. Where the two balance lines cross is the centre of gravity. For complex equipment, the manufacturer's manual or engineering drawings often show the CG location and the correct rigging attachment points.

#55 · Rigging Hoisting and Moving

What is load stability and why does a wide base improve it?

Answer

Load stability refers to how resistant a suspended load is to tipping or rotating. A load is more stable when the rigging attachment point is directly above the centre of gravity. A wider base of sling attachment points lowers the load's tendency to swing or tip because any tilt increases tension in the sling on that side, pulling it back.

#56 · Rigging Hoisting and Moving

What inspection must you do on a wire rope clip (Crosby clip) installation?

Answer

Check that the clips are installed with the U-bolt on the dead (short) end and the saddle on the live (load-bearing) end: never saddle a dead horse. Verify the correct number of clips for the rope diameter (minimum 3 for most sizes). Retighten clips after the first 10% of rated load is applied because the rope compresses slightly.

#57 · Rigging Hoisting and Moving

What is the rule for wire rope minimum bend radius around a sheave?

Answer

The minimum sheave diameter is typically 20 to 34 times the rope diameter, depending on the rope construction and the manufacturer's recommendation. Bending a wire rope around too small a sheave accelerates fatigue and causes individual wires to break faster. Always consult the rope manufacturer's data for the specific construction being used.

#58 · Rigging Hoisting and Moving

What does a load test on a newly installed overhead crane involve?

Answer

A load test applies 100% to 125% of the rated load to verify the crane structure, hoist, and brakes perform correctly before the crane is put into service. The test load is lifted, held stationary, and the brakes are checked. A qualified engineer must witness and sign off the test results.

#59 · Rigging Hoisting and Moving

Why should you never stand under a suspended load?

Answer

A suspended load can fall if the rigging fails, the crane loses power, or the load shifts. No one has time to move clear if a failure happens at height. Exclusion zones under and around crane lifts must be enforced, and the area must be barricaded to keep unauthorized workers out.

#60 · Rigging Hoisting and Moving

What does the term two-blocking mean on a crane and why is it dangerous?

Answer

Two-blocking occurs when the hook block travels upward until it contacts the boom tip sheave or structural member. The wire rope then goes tight between two fixed points and the hoist motor continues pulling, causing the rope or rigging hardware to fail suddenly. Modern cranes have two-block limit switches to prevent this, but they must be tested regularly.

#61 · Mechanical Power Transmission

What is the difference between a ball bearing and a cylindrical roller bearing?

Answer

A ball bearing uses spherical balls as rolling elements and can carry both radial and axial (thrust) loads. A cylindrical roller bearing uses cylinders, giving it higher radial load capacity but limited axial load capability. Ball bearings are used at higher speeds while cylindrical rollers are used for heavier radial loads.

#62 · Mechanical Power Transmission

What is a tapered roller bearing and what loads can it carry?

Answer

A tapered roller bearing uses cone-shaped rollers running between a tapered inner race (cone) and outer race (cup). It can carry heavy combined radial and one-direction axial (thrust) loads simultaneously. Tapered roller bearings are always installed in opposing pairs to handle thrust in both directions, such as in vehicle wheel hubs and gearboxes.

#63 · Mechanical Power Transmission

What is a spherical roller bearing and when is it preferred?

Answer

A spherical roller bearing has barrel-shaped rollers and a curved outer raceway that allows the inner ring and shaft to tilt slightly relative to the outer ring. This self-aligning feature makes it ideal for applications with shaft deflection, housing misalignment, or difficult-to-align long shafts like conveyor drives and paper mill rolls.

#64 · Mechanical Power Transmission

What is a thrust bearing and what type of load does it carry?

Answer

A thrust bearing is designed to carry purely axial (along the shaft axis) loads, with minimal or no radial load capacity. Ball thrust bearings, tapered roller thrust bearings, and spherical roller thrust bearings are common types. They are used in vertical pumps, screw conveyors, and anywhere the shaft pushes or pulls along its own axis.

#65 · Mechanical Power Transmission

How do you decode the bearing designation 6312?

Answer

In the ISO bearing numbering system, 6 means deep groove ball bearing, 3 means the bearing series (medium cross-section), and 12 is the bore code. Bore code 12 means multiply by 5: $12 \times 5 = 60$ mm bore. So bearing 6312 is a deep groove ball bearing with a 60 mm bore in the medium series.

#66 · Mechanical Power Transmission

What does the bore code 00, 01, 02, 03 mean in ISO bearing numbering?

Answer

For bore codes 00 through 03, the bore size does not follow the multiply-by-5 rule. Instead: 00 = 10 mm, 01 = 12 mm, 02 = 15 mm, 03 = 17 mm. From bore code 04 onward, you multiply by 5 to get the bore in millimetres ($04 \times 5 = 20$ mm, etc.).

#67 · Mechanical Power Transmission

What is the difference between an interference fit and a sliding fit for bearing mounting?

Answer

An interference fit for the inner ring on a shaft means the bearing bore is slightly smaller than the shaft diameter, so the ring grips the shaft tightly and rotates with it. A sliding (clearance) fit for the outer ring in a non-locating housing allows the outer ring to move axially as the shaft expands with heat. Misapplying these fits causes inner ring creep or locked thermal expansion.

#68 · Mechanical Power Transmission

What is bearing preload and when is it applied?

Answer

Preload is a controlled axial force applied to a bearing (or pair of bearings) to eliminate internal clearance and increase rigidity. It is used in precision spindles, machine tool axes, and angular contact bearing pairs to improve positional accuracy and reduce vibration. Too much preload increases heat and shortens bearing life.

#69 · Mechanical Power Transmission

What is bearing end float (axial play) and how do you check it?

Answer

End float is the total axial movement of a shaft when pushed from both ends in a bearing arrangement. You check it with a dial indicator on the shaft end while pushing the shaft back and forth axially. Correct end float (typically 0.05 to 0.25 mm for roller bearings) prevents over-constraint when the shaft expands with heat.

#70 · Mechanical Power Transmission

What is the difference between NLGI Grade 1 and NLGI Grade 2 bearing grease?

Answer

NLGI grades describe grease consistency: Grade 1 is softer, similar to tomato paste or soft shortening, and flows more easily, making it suitable for cold climates or slow speeds where pumpability matters. Grade 2 is firmer, roughly like peanut butter, and is the most common all-purpose bearing grease used in standard temperature and speed conditions. Grade 3 is even stiffer and is used in high-speed or high-temperature applications.

#71 · Mechanical Power Transmission

When should you use circulating oil lubrication instead of grease for bearings?

Answer

Use circulating oil when bearing speeds exceed the grease's speed rating, when continuous heat removal is needed (oil absorbs and carries away heat), or when the bearing is in a gearbox already filled with oil. Oil lubrication is also used for very large bearings or precision applications where the exact oil film thickness must be controlled.

#72 · Mechanical Power Transmission

What is a lip seal and what does it protect against?

Answer

A lip seal (also called a radial shaft seal or oil seal) has a flexible rubber lip held against a rotating shaft by a garter spring. It prevents lubricant from leaking out of a housing and stops contaminants like water, dust, and dirt from entering. The lip must contact the shaft on a smooth, undamaged surface; a worn groove on the shaft will cause leakage.

#73 · Mechanical Power Transmission

What is a labyrinth seal and where is it used?

Answer

A labyrinth seal is a non-contact seal made of interlocking rings or grooves that create a long, tortuous path. Contaminants and lubricants must travel this winding path to leak, which they generally cannot do at normal pressure. Labyrinth seals are used in high-speed, high-temperature applications (like turbines and large motors) where contact seals would wear too quickly.

#74 · Mechanical Power Transmission

What is a V-ring seal and how does it work?

Answer

A V-ring seal is a rubber ring that mounts on the shaft and rotates with it. Its flexible V-shaped lip bears axially against a stationary surface (like a housing or bearing outer ring), slinging contaminants outward by centrifugal force. It is simple, cheap, and effective for excluding dust and light splashing but is not suitable for high pressure.

#75 · Mechanical Power Transmission

What is a mechanical (face) seal and where is it used?

Answer

A mechanical seal uses two very flat, precision-lapped faces pressed together by a spring: one face rotates with the shaft and one is stationary. It seals high-pressure liquids (like in pumps handling hot water, chemicals, or hydrocarbons) where a lip seal would leak or fail quickly. Flush and cooling lines are often needed to lubricate and cool the sealing faces.

#76 · Mechanical Power Transmission

What are the two types of shaft misalignment a millwright must correct?

Answer

Angular misalignment means the two shaft centrelines meet at an angle (like two shafts pointing slightly toward each other). Parallel (offset) misalignment means the shaft centrelines are parallel but not co-linear (like two rails running side by side). Real machines usually have both types at the same time, and both must be corrected during alignment.

#77 · Mechanical Power Transmission

Describe the reverse dial indicator alignment method.

Answer

In the reverse dial method, two dial indicators are mounted on brackets: one indicator measures the other shaft's rim. You rotate both shafts together by 90-degree increments (0, 90, 180, 270 degrees) and record the readings. The difference between the 0-degree and 180-degree readings divided by 2 gives the offset at that bracket location. Geometry formulas then calculate the required shim changes and horizontal moves.

#78 · Mechanical Power Transmission

What are the typical laser alignment tolerances for a flexible coupling at 1,800 RPM?

Answer

At 1,800 RPM, a common target is parallel (offset) misalignment within 0.05 mm (0.002 inch) and angular misalignment within 0.05 mm per 100 mm of coupling span. Tighter tolerances (0.025 mm) are used for precision machines or couplings with limited flexibility. Always use the coupling manufacturer's alignment specification, which takes precedence over generic values.

#79 · Mechanical Power Transmission

What is a jaw (spider) coupling and what does it do for alignment?

Answer

A jaw coupling has two metal hubs with interlocking jaws and a rubber or polyurethane spider insert between them. The spider absorbs shock loads and tolerates small amounts of angular and parallel misalignment. It is not intended as an alignment substitute: the machine should still be aligned within specification before relying on the coupling's flexibility.

#80 · Mechanical Power Transmission

What is a disc coupling and what is its advantage over a gear coupling?

Answer

A disc coupling uses thin flexible metal discs (like washers) to transmit torque while accommodating misalignment through disc flexing. It has no wearing parts and needs no lubrication, making it maintenance-free and suitable for clean environments. A gear coupling uses meshing gear teeth that need periodic lubrication and will wear over time.

#81 · Mechanical Power Transmission

What is a gear coupling and when is it preferred over a jaw coupling?

Answer

A gear coupling uses internally and externally meshing crowned gear teeth to transmit very high torques while tolerating larger angular misalignment than a jaw coupling. It is preferred in heavy-duty rolling mill drives, steel plant equipment, and other applications with high torque and significant shaft deflection. It requires regular lubrication with EP gear grease or oil.

#82 · Mechanical Power Transmission

What is a fluid coupling and what does it provide that a rigid coupling cannot?

Answer

A fluid coupling transmits torque through a circulating oil vortex between an impeller (driver) and a runner (driven), with no mechanical connection between them. This provides a smooth, slip-limited start that protects the motor from shock loads and allows the motor to reach full speed before the driven machine accelerates. It acts as a torque limiter and cannot transmit more than the fluid vortex can support.

#83 · Mechanical Power Transmission

What is the difference between a classical V-belt (A, B, C section) and a narrow V-belt (3V, 5V, 8V)?

Answer

Classical V-belts have a wider, more open cross-section with a shallower wedge angle, and are the traditional industrial standard. Narrow V-belts have a taller, narrower profile that wedges more deeply into the sheave groove, transmitting more power per unit width. Narrow belts are more efficient, run cooler, and can replace multiple classical belts with fewer narrow belts.

#84 · Mechanical Power Transmission

How do you check V-belt tension using the deflection method?

Answer

Apply a perpendicular force at the midpoint of the belt span and measure how far the belt deflects. The required deflection is typically 16 mm (5/8 inch) per metre of span length, or about 1/64 inch per inch of span. Compare the force needed to produce that deflection to the manufacturer's recommended force range for that belt section and sheave speed.

#85 · Mechanical Power Transmission

What happens if a V-belt drive is over-tensioned?

Answer

Over-tensioning places excessive radial load on the shaft bearings, causing premature bearing failure. The belts themselves also wear faster, run hot, and can crack. Shaft deflection from the extra load can also cause misalignment. Over-tensioning is as harmful as under-tensioning and is more common because technicians tend to over-correct when belts squeal.

#86 · Mechanical Power Transmission

What is sheave groove wear and how does it affect belt drive performance?

Answer

As a V-belt wears, it rides lower in the sheave groove. When the groove sidewalls are also worn smooth (polished or undercut), the belt loses its wedging action and slips at lower torque loads than the drive was designed for. A worn groove can cause a new belt to bottom out in the groove (sitting on the groove floor instead of the sidewalls), eliminating the wedging effect entirely.

#87 · Mechanical Power Transmission

What is a synchronous (timing) belt and how does it differ from a V-belt?

Answer

A synchronous belt has teeth that mesh with grooves in matching sprockets, providing a positive (no-slip) drive. Unlike a V-belt, there is no wedging action and very low tension is needed, which reduces bearing loads. Synchronous belts are used where exact speed ratio and shaft synchronization are required, such as camshaft drives and servo systems.

#88 · Mechanical Power Transmission

What is the pitch of a synchronous belt and why does it matter?

Answer

Pitch is the distance from the centre of one belt tooth to the centre of the next, measured along the pitch line. The belt and sprocket must have exactly the same pitch, or the teeth will not mesh properly and the drive will skip or jam. Common pitches are identified by codes like 3M, 5M, 8M, or by older inch designations like L (9.525 mm), H (12.7 mm), and XH (22.225 mm).

#89 · Mechanical Power Transmission

What is chain sag and what is the correct amount for a roller chain drive?

Answer

Chain sag is the amount the slack strand of a chain drive droops under its own weight between the two sprockets. For a horizontal drive, the recommended sag is about 2% of the centre distance (the distance between the two sprocket centres). Too little sag means the chain is over-tight; too much means the chain is too loose and may skip sprocket teeth.

#90 · Mechanical Power Transmission

What is the difference between a simplex and duplex roller chain?

Answer

A simplex chain is a single-strand chain (one row of rollers and links), the most common type for standard power transmission. A duplex chain has two rows of rollers side by side, which roughly doubles the load capacity in the same sprocket diameter. Triplex (three strands) and higher-strand counts are available for even higher loads or where space limits sprocket size.

#91 · Mechanical Power Transmission

What causes roller chain stretch and when should the chain be replaced?

Answer

Chain stretch is actually wear elongation: the pins and bushings wear, making each link slightly longer. A chain is considered worn out when it has elongated by 3% or more (for most industrial chains). To measure wear, count 25 links (24 pitches) and compare the measured length to the nominal 24-pitch length. If the measured length is 3% over, replace the chain.

#92 · Mechanical Power Transmission

What is a spur gear and what is its main limitation?

Answer

A spur gear has straight teeth cut parallel to the shaft axis. It is the simplest and least expensive gear type, used for low-to-moderate speed parallel shaft drives. The main limitation is noise: all teeth engage along their full width at the same instant, causing impact loading and a characteristic whining or thumping noise at higher speeds.

#93 · Mechanical Power Transmission

What is a helical gear and what advantage does it have over a spur gear?

Answer

A helical gear has teeth cut at an angle (the helix angle) to the shaft axis. Because the teeth engage gradually along their length rather than all at once, helical gears run more smoothly and quietly than spur gears at the same speed. The trade-off is an axial thrust force that requires thrust bearings to absorb.

#94 · Mechanical Power Transmission

What is a bevel gear and what shaft arrangement does it connect?

Answer

A bevel gear has teeth cut on a conical surface and is used to transmit power between shafts whose axes intersect, most commonly at a 90-degree angle. Straight bevel gears are simple but noisy. Spiral bevel gears have curved teeth that engage more gradually, running quieter and carrying higher loads, similar to the helical vs spur comparison.

#95 · Mechanical Power Transmission

What is a worm gear set and what are its characteristics?

Answer

A worm gear set consists of a threaded worm (like a screw) meshing with a worm wheel (a helical gear). It can produce very high reduction ratios (10:1 to 100:1 or more) in a compact space with smooth, quiet operation. The main limitations are low efficiency (typically 40-90%) and self-locking tendency at high reduction ratios, where the worm can drive the wheel but not the reverse.

#96 · Mechanical Power Transmission

How do you calculate the gear ratio of a two-gear mesh?

Answer

Gear ratio equals the number of teeth on the driven gear divided by the number of teeth on the driving gear. For example, if a 20-tooth pinion drives a 60-tooth gear, the ratio is $60 \div 20 = 3:1$, meaning the output shaft turns once for every three turns of the input shaft. Speed ratio is the inverse: output speed equals input speed divided by the ratio.

#97 · Mechanical Power Transmission

What is gear module and how is it used in gear design?

Answer

Module is the ratio of the pitch circle diameter to the number of teeth ($m = d/N$), measured in millimetres. It defines the size of the gear tooth: a higher module means larger, stronger teeth. Two gears can only mesh if they have exactly the same module. Module is the metric equivalent of diametral pitch used in the inch system.

#98 · Mechanical Power Transmission

What is diametral pitch (DP) in gear terminology?

Answer

Diametral pitch is the number of gear teeth per inch of pitch circle diameter ($DP = N/d$ in inches). It is the inverse of module and is used in the inch gear system. A higher DP number means smaller, finer teeth; a lower DP means larger teeth. Two meshing gears must have the same DP, just as metric gears must share the same module.

#99 · Mechanical Power Transmission

What is the addendum of a gear tooth?

Answer

The addendum is the height of the gear tooth above the pitch circle (the imaginary circle where two meshing gears contact). It equals one module for a standard gear (or $1/DP$ in inch units). The addendum determines how far the tooth tip projects beyond the pitch circle and affects how much the teeth overlap during meshing.

#100 · Mechanical Power Transmission

What is the dedendum of a gear tooth?

Answer

The dedendum is the depth of the gear tooth below the pitch circle, down to the root (base) of the tooth. It is slightly larger than the addendum (typically $1.25 \times$ module) to provide clearance so the tip of the mating gear tooth does not bottom out in the root. The total tooth height is addendum plus dedendum.

#101 · Mechanical Power Transmission

What is gear backlash and how do you measure it?

Answer

Backlash is the clearance between the non-driving flanks of mating gear teeth, measured along the pitch circle. It is needed to prevent gears from binding due to thermal expansion or machining tolerances. Measure it by holding one gear fixed and using a dial indicator to measure the rotational movement of the mating gear at its pitch circle before the flanks engage.

#102 · Mechanical Power Transmission

What type of oil is typically used in an industrial worm gearbox?

Answer

Worm gearboxes require high-viscosity gear oil with extreme pressure (EP) and anti-wear additives, often a compounded oil or synthetic polyglycol (PAG) fluid. The high sliding action between the worm and wheel generates more heat and boundary lubrication conditions than helical or spur gears. PAG oils can reduce worm gear operating temperature by 20-30 degrees Celsius compared to mineral EP oils.

#103 · Mechanical Power Transmission

How often should industrial gearbox oil be changed and what triggers an early change?

Answer

A common interval is annually for mineral oil or every 2-3 years for synthetic oil under normal conditions. Early change is triggered by oil analysis showing high water content, metal particle counts above baseline, viscosity out of spec, or acid number increase. Never mix oil types or brands without consulting the gearbox manufacturer, as some additive packages are incompatible.

#104 · Mechanical Power Transmission

What is the purpose of a breather on a gearbox?

Answer

A gearbox breather (vent plug) allows air to flow in and out as the gearbox heats up and cools down, preventing pressure buildup inside the housing. Without a breather, pressure differentials can force oil past seals. The breather must have a filter element to prevent dust and moisture from entering with the incoming air.

#105 · Mechanical Power Transmission

What is coupling alignment soft foot and how does it cause problems?

Answer

Soft foot is a condition where one or more machine feet do not make solid, flat contact with the base plate when the hold-down bolts are loose. When you tighten the bolts, the frame distorts, which misaligns the shaft and causes the bearing housing to distort internally. Soft foot must be corrected with shims before any alignment work begins.

#106 · Mechanical Power Transmission

How do you check for soft foot before aligning a machine?

Answer

Mount a dial indicator on the machine frame near each foot. Loosen one foot bolt while all others are tight and watch the dial indicator. If the indicator moves more than 0.05 mm (0.002 inch) when that bolt is loosened, soft foot is present at that foot. Shim the gap with precision stainless shims until the movement is within tolerance, then recheck all feet.

#107 · Mechanical Power Transmission

What is thermal growth and how does it affect machine alignment?

Answer

When a machine runs, heat causes the metal to expand, raising the shaft centreline height above its cold position. If you align the machine cold, it will be misaligned when hot. Thermal growth corrections are calculated from the machine's operating temperature, material coefficient of thermal expansion, and the height from the base to the shaft centreline, then built into the cold alignment target as an offset.

#108 · Mechanical Power Transmission

What is a press fit and how is it accomplished for a large bearing inner ring?

Answer

A press fit means the bearing bore is smaller than the shaft diameter, creating an interference. For small bearings, a mechanical press and sleeve are used to push the inner ring onto the shaft. For larger bearings, the ring is heated (by induction heater or oil bath, typically 80-100 degrees Celsius above ambient) so it expands enough to slip onto the shaft, then is pushed to its seat before it cools and grips.

#109 • Mechanical Power Transmission

What is the maximum temperature for heating a bearing inner ring during installation?

Answer

Heat the inner ring to between 80 and 100 degrees Celsius (total temperature of the ring, not rise above ambient). Never allow bearing steel to exceed 120 to 125 degrees Celsius, as higher temperatures can reduce hardness through tempering and permanently damage the bearing. Never use an open flame; use an induction heater, oil bath, or hot plate so the temperature is controlled and measured.

#110 • Mechanical Power Transmission

What is the difference between grease re-lubrication interval and quantity?

Answer

Re-lubrication interval is how often you add grease (based on bearing size, speed, temperature, and environment). Re-lubrication quantity is how much grease to add each time, calculated from the formula: $G = 0.005 \times D \times B$, where D is bearing outside diameter in mm and B is total bearing width in mm, giving grams of grease. Over-greasing causes churning, heat, and premature bearing failure.

#111 · Mechanical Power Transmission

What does bearing L10 life mean?

Answer

L10 (or B10) life is the number of operating hours or revolutions by which 10% of a large group of identical bearings under the same conditions will have failed due to fatigue. In other words, 90% of bearings are expected to survive to the L10 life. It is the standard basis for bearing selection and is calculated using the bearing's dynamic load rating and the applied load.

#112 · Mechanical Power Transmission

What is a V-belt drive speed ratio formula?

Answer

Speed ratio equals the pitch diameter of the driven sheave divided by the pitch diameter of the driver sheave. For example, a 150 mm driver sheave turning at 1,750 RPM connected to a 450 mm driven sheave gives a ratio of 3:1, so the driven shaft turns at 1,750 divided by 3 = 583 RPM. Use pitch diameters (not outside diameters) for accurate calculations.

#113 · Mechanical Power Transmission

What is the contact angle in a V-belt drive and how does a small driven sheave affect it?

Answer

Contact angle (also called arc of contact) is the angle over which the belt wraps around the sheave. The contact angle is 180 degrees on both sheaves only when both sheaves are exactly the same diameter (a 1:1 speed ratio). When the driven sheave is smaller than the driver, the belt wraps less of the small sheave, reducing grip and lowering the power the drive can transmit before the belt slips.

#114 · Mechanical Power Transmission

What is a matched set of V-belts and why is it required for multi-belt drives?

Answer

A matched set means all belts in a multi-groove drive are within a very tight length tolerance of each other (typically within 2-4 mm). If one belt is longer than the others, it hangs slack while the shorter belts carry all the load, causing uneven wear and overloading the short belts. Always replace a multi-belt drive with a complete new matched set, never just one belt.

#115 · Mechanical Power Transmission

What is the chain drive lubrication requirement for a high-speed drive?

Answer

High-speed roller chain drives (above about 600 RPM at the sprocket) require oil bath or oil stream (force-feed) lubrication, not manual or drip lubrication. The oil flushes heat, carries away debris, and provides a full film on the pin-bushing interface where wear occurs. Using grease on a high-speed chain drive causes overheating because grease cannot circulate to remove heat.

#116 · Mechanical Power Transmission

What is the pitch of a No. 50 roller chain?

Answer

The ANSI chain number system uses the first one or two digits to indicate pitch in eighths of an inch. No. 50 chain has a pitch of $\frac{5}{8}$ inch (about 15.875 mm). No. 40 chain has a $\frac{1}{2}$ inch pitch, No. 60 has a $\frac{3}{4}$ inch pitch, and No. 80 has a 1-inch pitch. The last digit (0 for standard, 1 for lightweight) indicates the series.

#117 · Mechanical Power Transmission

What is a sprocket pitch diameter and how do you calculate it for a No. 50 chain on a 17-tooth sprocket?

Answer

The pitch diameter formula is: $PD = P \div \sin(180 \text{ degrees} / N)$, where P is the chain pitch and N is the number of teeth. For a No. 50 chain ($P = 15.875 \text{ mm}$) on 17 teeth: $PD = 15.875 / \sin(180/17) = 15.875 / \sin(10.59 \text{ degrees}) = 15.875 / 0.1837 =$ approximately 86.4 mm. This is the diameter at which the chain pins are centred.

#118 · Mechanical Power Transmission

What causes pitting corrosion on gear teeth and how is it prevented?

Answer

Pitting is surface fatigue caused by repeated contact stress exceeding the surface fatigue limit of the gear material. Small craters form on the tooth flank in the pitch line area. It is prevented by using the correct oil viscosity and extreme-pressure additives to maintain an adequate oil film, avoiding overloading the gearbox, and selecting gears with appropriate surface hardness for the application.

#119 · Mechanical Power Transmission

What is gear scoring (scuffing) and what causes it?

Answer

Scoring (or scuffing) is severe adhesive wear on gear tooth flanks, appearing as scratches or grooves running in the direction of tooth sliding. It is caused by oil film breakdown under high load or high speed, allowing metal-to-metal contact and micro-welding. Prevention requires the correct EP gear oil viscosity grade, adequate oil quantity, and not overloading the gearbox.

#120 · Mechanical Power Transmission

What is the pressure angle of a standard modern gear tooth and what does it mean?

Answer

The standard pressure angle for modern gears is 20 degrees (older designs used 14.5 degrees). The pressure angle is the angle between the line of action (the direction the tooth force acts) and the tangent to the pitch circle at the contact point. A 20-degree pressure angle gives stronger, wider teeth at the root compared to 14.5 degrees, allowing higher torque transmission.

#121 · Mechanical Power Transmission

How do you determine if a gearbox has the correct oil level during maintenance?

Answer

Most gearboxes have a sight glass or dipstick on the side of the housing. The oil level should be at the centre of the sight glass or within the min-max marks on the dipstick when the unit is at rest and the oil has settled (typically 10-15 minutes after shutdown). Checking level on a running gearbox gives a false low reading because oil is circulating inside.

#122 · Mechanical Power Transmission

What is a flexible disc coupling's failure mode and how do you inspect it?

Answer

The flexible discs fatigue and develop cracks or missing sections after long service or if the drive is operated with excessive misalignment. Inspect by rotating the coupling by hand with the power locked out and visually examining each disc pack for cracks, missing bolts, fretting between the disc layers, and corrosion. Replace the disc pack if any crack or bolt hole elongation is found.

#123 • Mechanical Power Transmission

What is the purpose of an elastomeric insert in a jaw coupling?

Answer

The elastomeric (rubber or polyurethane) spider insert fills the gap between the interlocking jaws of the two coupling hubs. It transmits torque, absorbs shock and vibration, and allows small amounts of misalignment. When the spider is worn, cracked, or missing pieces, metal-to-metal contact between the jaws causes noise, vibration, and rapid hub wear. Inspect the spider regularly through the coupling guard window if available.

#124 • Mechanical Power Transmission

What is a keyway and key, and why must they be inspected during a coupling changeout?

Answer

A keyway is a rectangular slot machined into both the shaft and the hub bore. A key (a rectangular bar) fits into the aligned slots to prevent the hub from rotating on the shaft. During a coupling changeout, inspect the key and keyway for brinelling (indentations), cracks, rounding of corners, and fretting corrosion. A damaged key or battered keyway will allow the hub to rock and fatigue the shaft.

#125 · Mechanical Power Transmission

What is a taper-lock (taper-bush) hub and what is its advantage?

Answer

A taper-lock bush is a split tapered sleeve that fits inside a matching taper in a sprocket, sheave, or coupling hub. Tightening the flange bolts draws the bush into the taper, gripping the shaft without a keyway (or with one for added security). It is quick to install and remove, self-centering, and distributes clamping force evenly around the shaft, making it less likely to crack the hub than a standard interference fit.

#126 · Mechanical Power Transmission

What is the purpose of a keyway in a shaft assembly?

Answer

A keyway is a slot cut into a shaft and a hub so a key can be inserted to lock them together and transmit torque. Without a key, the hub would spin freely on the shaft. The key prevents relative rotation between the two parts.

#127 · Mechanical Power Transmission

What is a square key and when is it used?

Answer

A square key has equal width and height and sits half in the shaft and half in the hub. It is the most common key type for general-purpose torque transmission. Square keys are used when loads are moderate and the fit can be controlled closely.

#128 · Mechanical Power Transmission

What is a Woodruff key and what makes it different from a square key?

Answer

A Woodruff key is a semicircular disc that seats in a curved keyseat milled into the shaft. It self-aligns and is good for tapered shaft ends where axial loads are present. The curved shape means it cannot be driven out of the shaft as easily as a flat key.

#129 · Mechanical Power Transmission

What is a splined shaft and when is it preferred over a keyed shaft?

Answer

A splined shaft has multiple teeth machined around its circumference that mesh with matching grooves in the hub. Splines distribute torque over a larger contact area than a single key, so they handle higher loads. They also allow the hub to slide axially along the shaft under load.

#130 · Mechanical Power Transmission

How do you inspect a keyseat for wear before reassembly?

Answer

Check the keyseat walls for rounding, broaching, or galling caused by a loose key rocking back and forth. Use a new key to check the fit; there should be no side play. If the keyseat is damaged, the shaft or hub must be remachined or replaced.

#131 • Mechanical Power Transmission

What is an interference fit and why is it used on shafts?

Answer

An interference fit, also called a press fit, means the shaft diameter is slightly larger than the bore it goes into. The parts are forced together and the resulting compression holds them without fasteners. It is used when high torque must be transmitted and a key alone is not enough.

#132 • Mechanical Power Transmission

How do you calculate the interference required for a press fit?

Answer

Interference is the difference between the shaft outer diameter and the hub bore diameter. The required value comes from the material properties, the torque to be transmitted, and the fit class specified on the drawing (such as FN2 or H7/p6). A millwright uses the engineer's specification or a standard table to find the correct amount.

#133 • Mechanical Power Transmission

What is a shrink fit and how does it differ from a cold press fit?

Answer

A shrink fit uses heat to expand the hub so the shaft slides in easily, then the hub contracts as it cools to grip the shaft tightly. A cold press fit forces the parts together at room temperature using a hydraulic press. Shrink fits create less stress on the parts and are preferred for large or fragile components.

#134 • Mechanical Power Transmission

What temperature is typically used to heat a hub for a shrink fit?

Answer

Most steel hubs are heated to between 100 and 150 degrees Celsius to achieve enough expansion to slide over the shaft. Heavy interference fits on large bores may require up to 175 degrees Celsius, but this is the practical upper limit for most steels. Overheating changes the metal's properties, so a temperature-indicating crayon or infrared thermometer is used to verify the temperature.

#135 · Mechanical Power Transmission

What is shaft runout and how is it measured?

Answer

Shaft runout is how much a shaft wobbles or deviates from its true center as it rotates. It is measured by placing a dial indicator against the shaft surface and turning the shaft slowly by hand. The total indicator reading (TIR) is the difference between the highest and lowest readings on the dial.

#136 · Mechanical Power Transmission

What is the difference between TIR and actual shaft deviation?

Answer

TIR stands for Total Indicator Reading, which is the full swing of the dial indicator from its lowest to its highest reading. The actual eccentricity, meaning how far the shaft centerline is from true center, is half the TIR value. A TIR of 0.10 mm means the shaft center is only 0.05 mm off true.

#137 · Mechanical Power Transmission

What causes excessive shaft runout in a rotating machine?

Answer

Common causes include a bent shaft from an impact or heat, worn or damaged bearings that allow the shaft to wobble, and a component such as a gear or pulley that is mounted out of round. Improper machining of the shaft journal or a loose fit in the bearing housing can also produce runout.

#138 · Mechanical Power Transmission

What is a flexible coupling and why is it used?

Answer

A flexible coupling connects two shafts and allows for small amounts of misalignment, both angular and parallel. It absorbs shock loads and vibration so they do not pass directly from one shaft to the other. Flexible couplings protect bearings and gears from damage caused by alignment errors or sudden load changes.

#139 · Mechanical Power Transmission

What is a rigid coupling and when should it be used?

Answer

A rigid coupling connects two shafts with no flexibility, meaning both shafts must be perfectly aligned before the coupling is installed. It is used when shafts are very precisely aligned by design, such as in precision machine tools. If misalignment exists with a rigid coupling, bearing and shaft damage will follow quickly.

#140 · Mechanical Power Transmission

What are the main types of flexible coupling elements?

Answer

Common flexible elements include rubber or polyurethane spiders used in jaw couplings, thin stainless steel disc packs used in disc couplings, and rubber tires used in tire couplings. Some couplings use grid springs or gear teeth to allow movement. The flexible element wears over time and must be inspected and replaced on a schedule.

#141 • Mechanical Power Transmission

How do you inspect a jaw coupling spider element?

Answer

Remove the coupling hubs and pull out the rubber or urethane spider. Check it for cracks, chunking, permanent compression, or hardening. A damaged spider allows the two coupling halves to contact each other metal-to-metal, which will quickly damage shafts and bearings.

#142 • Mechanical Power Transmission

What is a disk stack coupling and what is its angular limit?

Answer

A disk stack coupling uses a pack of thin flexible metal discs bolted alternately to each coupling hub to transmit torque while allowing small misalignment. The angular misalignment limit is typically 1 degree or less per disc pack, depending on the manufacturer. Exceeding this limit fatigues the discs and causes cracking.

#143 · Mechanical Power Transmission

How do you inspect a flexible disc coupling for damage?

Answer

Look at each disc in the pack for cracks, especially near the bolt holes where stress concentrates. Check the bolts and bushings for wear, looseness, or shearing. A cracked disc must be replaced as a complete pack, not as individual discs, to keep the load balanced across all discs.

#144 · Mechanical Power Transmission

What is gear backlash and why does it matter?

Answer

Gear backlash is the small gap between the non-driving faces of two meshing gear teeth. Some backlash is needed to allow for lubrication and thermal expansion. Too much backlash causes impact loading and noise when direction reverses; too little causes the gears to bind and overheat.

#145 · Mechanical Power Transmission

How do you measure gear backlash with a dial indicator?

Answer

Lock one gear in place and place a dial indicator tip against a tooth face of the mating gear. Rock the free gear back and forth gently and read the total movement on the indicator. Compare the reading to the manufacturer's specification, which is typically stated in thousandths of an inch or hundredths of a millimeter.

#146 · Mechanical Power Transmission

What causes gear backlash to increase over time?

Answer

Normal wear on the tooth flanks removes material and widens the gap between meshing teeth. Poor lubrication speeds up this wear significantly. Excessive backlash is a sign the gear set is nearing end of life and should be monitored closely or scheduled for replacement.

#147 · Mechanical Power Transmission

What is a timing belt and how does tension affect its performance?

Answer

A timing belt has teeth molded into its inner surface that mesh with matching grooves in the pulleys to maintain a fixed speed ratio without slipping. If tension is too low, the belt can skip teeth, which causes timing errors. If tension is too high, it overloads the bearings and shortens belt life.

#148 · Mechanical Power Transmission

How do you adjust a timing belt tensioner correctly?

Answer

Use a belt tension gauge or the manufacturer's deflection method, which specifies how much the belt should deflect when a set force is applied at the midspan. Set the tensioner to achieve that exact deflection, then lock it in place. Recheck tension after the first hour of operation because new belts seat and tension drops.

#149 · Mechanical Power Transmission

What is a sheave and how do you inspect it for wear?

Answer

A sheave is the grooved wheel that a V-belt or timing belt rides in. Inspect the grooves for wear by checking if the belt sits too low in the groove, which means the sides of the groove have worn away. A worn sheave must be replaced because a belt riding on the groove bottom loses its wedging action and slips.

#150 · Mechanical Power Transmission

What is sheave groove angle and why must it match the belt?

Answer

The groove angle is the angle of the V-shape cut into the sheave. It must match the belt's angle so the belt contacts both groove walls evenly and uses the wedging effect to transmit torque. A mismatched angle causes the belt to bottom out or ride too high, both of which reduce grip and accelerate wear.

#151 · Mechanical Power Transmission

What is a variable speed mechanical drive and how does it work?

Answer

A variable speed mechanical drive changes the output speed by adjusting the effective diameter of the sheaves or pulleys while the belt is running. One common type uses movable sheave flanges that open or close to change the pitch diameter the belt rides on. As one sheave opens, the belt drops lower and the ratio changes.

#152 · Mechanical Power Transmission

What maintenance does a variable speed mechanical drive require?

Answer

Keep the movable sheave flanges clean and lightly lubricated so they slide smoothly. Check the belt for wear caused by the constant change in pitch radius. Inspect the adjustment mechanism for wear, and verify the speed range is still within specification using a tachometer.

#153 · Mechanical Power Transmission

What is an overrunning clutch and where is it used?

Answer

An overrunning clutch, also called a sprag or roller ramp clutch, transmits torque in one direction but freewheels when the driven shaft tries to rotate faster than the driving shaft. It is used on conveyors to prevent reverse motion and on backstop applications to stop a loaded incline conveyor from running backward.

#154 · Mechanical Power Transmission

How do you check an overrunning clutch for proper operation?

Answer

Rotate the clutch in the driving direction and confirm it locks up and transmits torque. Then rotate it in the freewheeling direction and confirm it spins freely with no drag or ratcheting noise. Any roughness or hesitation in the freewheeling direction means the sprags or rollers are worn and the unit needs rebuilding.

#155 · Mechanical Power Transmission

What is a backstop device on a conveyor?

Answer

A backstop is a mechanical device, usually a sprag or roller clutch built into the drive pulley or gearbox, that allows the conveyor to run forward but locks automatically if the belt tries to run backward. It is required on inclined conveyors so the loaded belt does not roll back when the drive stops.

#156 · Mechanical Power Transmission

How do you inspect and test a backstop after installation?

Answer

Confirm the backstop is installed with the locking direction opposing the loaded rollback direction. Try to rotate the shaft manually in the rollback direction; it should lock solidly. Run the conveyor and then stop it under load to confirm the backstop holds without slipping or chatter.

#157 · Mechanical Power Transmission

What is a torque limiter and what problem does it prevent?

Answer

A torque limiter is a coupling device that slips or disconnects when the torque exceeds a set value, protecting the driven machine from overload damage. It acts like a mechanical fuse between the motor and the machine. Without it, a jam or sudden shock load could bend shafts, break gears, or damage the gearbox.

#158 · Mechanical Power Transmission

How do you set the slip torque on a friction-style torque limiter?

Answer

Consult the manufacturer's chart to find the correct spring compression or clamping force for the desired torque value. Adjust the compression nut to that setting, then test by applying a known torque and confirming the unit slips at the right value. Recheck the setting after the first few months of service because friction pads seat and the torque level can drop.

#159 · Mechanical Power Transmission

What is a universal joint and where is it commonly used?

Answer

A universal joint, also called a U-joint or Cardan joint, connects two shafts that meet at an angle and allows torque to be transmitted even as the angle changes. It is used on drive shafts in vehicles, on paper machine rolls, and on any application where the shaft angle must flex during operation.

#160 · Mechanical Power Transmission

What is universal joint phasing and why does it matter?

Answer

Phasing means the two yokes of a double U-joint driveshaft are aligned in the same plane. When phasing is correct, the speed fluctuation produced by the first joint is cancelled out by the second joint, giving smooth output speed. Incorrect phasing causes a pulsing torque and vibration at twice the rotation frequency.

#161 · Mechanical Power Transmission

How do you check U-joint phasing on a double-joint driveshaft?

Answer

Look along the shaft from one end. The ears (yokes) on both ends of the intermediate shaft must be in the same plane, meaning they line up visually. Most manufacturers mark the yokes with alignment marks. If the marks are 90 degrees off, the phasing is wrong and the shaft must be disassembled and reassembled correctly.

#162 · Mechanical Power Transmission

How do you inspect a universal joint for wear?

Answer

Check the cross and bearing cups by holding the yoke and trying to move the cross in every direction. Any looseness or play means the needle bearings inside the cup are worn. Worn U-joints cause vibration and can eventually fracture the cross, causing a sudden and dangerous driveshaft failure.

#163 · Mechanical Power Transmission

What is a spline fit inspection and what defects do you look for?

Answer

Slide the splined hub onto the splined shaft and check for excessive play, which means the spline teeth are worn. Look at the tooth flanks for fretting wear, which appears as a reddish brown powder, and for pitting or cracking. A worn spline connection will cause torsional slop and impact loading under reversing loads.

#164 · Mechanical Power Transmission

What lubrication does a sliding spline connection require?

Answer

Sliding splines need grease or oil to prevent fretting corrosion and reduce wear as the spline slides axially. The grease must be compatible with the operating temperature and should be specified by the manufacturer. Without lubrication, spline teeth corrode and seize, which prevents the axial movement the joint was designed to allow.

#165 · Mechanical Power Transmission

What is the angular misalignment limit for a flexible disc coupling?

Answer

Most single disc pack flexible couplings allow no more than 0.5 to 1 degree of angular misalignment, and the exact limit is on the manufacturer's datasheet. Exceeding this limit creates bending stress in the discs at every revolution. Over time this fatigue cracks the discs, and a failure can be sudden and violent.

#166 · Mechanical Power Transmission

What is the difference between angular and parallel shaft misalignment?

Answer

Angular misalignment means the two shaft centerlines meet at an angle rather than being truly parallel. Parallel misalignment, also called offset, means the shafts are parallel but their centerlines are not on the same line. Most flexible couplings must handle both types at the same time, and the combined misalignment must stay within the rating.

#167 · Mechanical Power Transmission

What tools are used to align shafts before coupling installation?

Answer

Traditional methods use dial indicators mounted on magnetic bases, with readings taken at multiple positions as the shaft rotates. Laser alignment systems use a laser transmitter on one shaft and a receiver on the other for faster and more accurate results. Both methods measure angular and parallel offset to tell the millwright how much to move the machine.

#168 · Mechanical Power Transmission

What is soft foot and how does it affect shaft alignment?

Answer

Soft foot means one or more of the machine's mounting feet do not sit flat on the base, so tightening the bolts distorts the machine frame and shifts the shaft position. You find it by loosening one foot bolt at a time while watching a dial indicator on the shaft; a large reading change means that foot is soft. Soft foot must be corrected with shims before alignment can be completed.

#169 · Mechanical Power Transmission

What is thermal growth and how does it affect shaft alignment?

Answer

Thermal growth is the increase in machine height that happens as the machine heats up from cold start to operating temperature. A pump driven by a hot motor may be at the correct alignment when cold but out of alignment when running. Millwrights compensate by intentionally offsetting the cold alignment by the calculated growth amount.

#170 · Mechanical Power Transmission

What is a gear coupling and what are its advantages?

Answer

A gear coupling uses two externally toothed hubs that mesh inside internally toothed sleeves to transmit torque. The tooth profile allows a small amount of angular and parallel misalignment while carrying very high torque. Gear couplings require periodic lubrication of the teeth and are common on heavy industrial drives.

#171 · Mechanical Power Transmission

How do you inspect a gear coupling for wear?

Answer

Disassemble the coupling and look at the teeth on both the hub and the sleeve for wear, pitting, and fretting. Check the tooth flanks for rounding, which indicates the coupling has been running with too much misalignment. Also inspect the O-rings or seals that keep grease inside the sleeve.

#172 · Mechanical Power Transmission

What is a jaw coupling and how does it transmit torque?

Answer

A jaw coupling has two metal hubs with claw-shaped jaws that interleave with each other. A rubber or polyurethane spider sits between the jaws and is compressed as torque is applied. The spider absorbs shock and damps vibration, making this coupling popular on small to medium pumps and compressors.

#173 • Mechanical Power Transmission

What is a grid coupling and how does it work?

Answer

A grid coupling has two flanged hubs with slots around their circumference. A single piece of spring steel grid weaves through the slots of both hubs to connect them. The grid flexes to allow misalignment and deflects under shock loads, absorbing energy the way a spring does.

#174 • Mechanical Power Transmission

What is a chain coupling and what are its limitations?

Answer

A chain coupling uses two sprockets, one on each shaft, connected by a short length of double-strand roller chain. It handles moderate misalignment and is easy to disassemble. However, the chain needs lubrication and wears over time, so chain couplings are less common where maintenance access is difficult.

#175 · Mechanical Power Transmission

What is torsional vibration in a drivetrain?

Answer

Torsional vibration is a twisting oscillation that travels along shafts and couplings when the torque applied to the system changes rapidly or at a resonant frequency. It can crack shafts and coupling components without producing obvious vibration that an operator can feel. It is detected with shaft-mounted strain gauge telemetry, torsional accelerometers, or encoder-based angular velocity measurement devices.

#176 · Mechanical Power Transmission

What is a brake and how does it differ from a clutch?

Answer

A brake stops or holds a rotating shaft by applying friction or clamping force to bring it to rest. A clutch connects or disconnects two rotating shafts to start or stop torque transmission. The difference is that a brake always acts against something stationary, while a clutch connects two parts that are both capable of rotating.

#177 · Mechanical Power Transmission

What is a disc brake and how do you inspect its pads?

Answer

A disc brake clamps friction pads against a rotating disc called a rotor to generate stopping force. Inspect the pad thickness and compare it to the manufacturer's minimum; worn pads must be replaced before they reach metal-to-metal contact. Also check the rotor surface for scoring, heat cracks, and minimum thickness.

#178 · Mechanical Power Transmission

What is a fail-safe brake and why is it used on hoists?

Answer

A fail-safe brake is spring-applied and held open by hydraulic or electric pressure when power is on. If power is lost for any reason, the spring automatically applies the brake and stops the load. Hoists and elevating conveyors use fail-safe brakes so a power failure does not allow a suspended load to fall.

#179 · Mechanical Power Transmission

What is a friction clutch and how is slip torque set?

Answer

A friction clutch uses spring-loaded friction discs or plates to transmit torque up to a maximum set value, then it slips to protect the drivetrain. The slip torque is set by adjusting the spring compression, with more compression giving a higher torque setting. After adjustment, test the clutch under load to confirm it slips at the correct torque.

#180 · Mechanical Power Transmission

What is a fluid coupling and what advantage does it offer on heavy starts?

Answer

A fluid coupling uses oil to transmit torque between an impeller on the driving side and a runner on the driven side, with no mechanical contact between them. Because the connection is through fluid, the motor can reach full speed before full torque is applied to the load, giving a very smooth start. This reduces mechanical shock and motor starting current on heavy loads like large conveyors.

#181 · Material Handling and Process Systems

What are the main parts of a centrifugal pump?

Answer

The main parts are the impeller, which spins and throws the liquid outward; the casing or volute, which collects the liquid and converts velocity into pressure; the wear rings, which reduce recirculation losses; and the stuffing box or seal chamber, which prevents liquid from leaking along the shaft. The shaft transmits power from the motor to the impeller.

#182 · Material Handling and Process Systems

What is the function of wear rings in a centrifugal pump?

Answer

Wear rings create a close clearance between the rotating impeller and the stationary casing to minimize the amount of high-pressure liquid that leaks back to the suction side. As the rings wear, this clearance grows, efficiency drops, and the pump must work harder to maintain flow. Wear rings are replaceable so the pump body does not have to be discarded when clearances open up.

#183 · Material Handling and Process Systems

What is a pump curve and what does it show?

Answer

A pump curve is a graph showing the relationship between flow rate on the horizontal axis and total head (pressure) on the vertical axis for a specific pump at a given speed. As flow increases, head decreases. The pump operates at the point where the pump curve intersects the system curve, which represents the resistance of the piping system.

#184 · Material Handling and Process Systems

What does the best efficiency point (BEP) mean on a pump curve?

Answer

The BEP is the flow rate at which the pump operates at its highest efficiency, meaning the least energy is wasted. Operating far from the BEP causes increased radial loads on the shaft, higher vibration, and accelerated wear on the seal and bearings. A pump should be selected so normal operating conditions are close to the BEP.

#185 · Material Handling and Process Systems

What is NPSH and why is it critical for pump operation?

Answer

NPSH stands for Net Positive Suction Head and is the amount of pressure available at the pump inlet above the vapor pressure of the liquid being pumped. NPSHa is the available suction head from the system, and NPSHr is the minimum the pump requires to function without cavitating. NPSHa must always be greater than NPSHr, with a safety margin, or the pump will cavitate.

#186 · Material Handling and Process Systems

What is cavitation in a pump and what damage does it cause?

Answer

Cavitation happens when the pressure at the impeller eye drops below the liquid's vapor pressure, forming vapor bubbles. These bubbles collapse violently as they move into higher-pressure zones, creating tiny explosions that erode the impeller and casing metal. Signs of cavitation include a crackling or gravel-like noise, reduced flow, and pitting damage on the impeller vanes.

#187 · Material Handling and Process Systems

What are common causes of pump cavitation in the field?

Answer

The most common causes are a suction line that is too long or too small, a suction screen or strainer that is partially plugged, a suction isolation valve that is not fully open, and pumping a hot liquid close to its boiling point. Running a pump above its design flow rate also reduces the inlet pressure below the required NPSHr.

#188 · Material Handling and Process Systems

What is pump priming and when is it required?

Answer

Priming means filling the pump casing and suction pipe with liquid before starting so the impeller has something to throw. Centrifugal pumps cannot pull liquid up from a lower level on their own if air is present because air is too light for the impeller to generate useful suction. Pumps below the liquid level (flooded suction) are self-priming; pumps above the liquid level need a foot valve, a vacuum priming system, or a self-priming pump design.

#189 · Material Handling and Process Systems

What is the difference between pump packing and a mechanical seal?

Answer

Packing is braided material compressed in the stuffing box around the shaft by a gland follower to restrict but not completely stop leakage; some drip is needed to lubricate and cool the packing. A mechanical seal uses two lapped flat faces, one rotating and one stationary, that seal against each other with spring pressure, allowing essentially zero leakage. Mechanical seals require less maintenance and are preferred for hazardous or environmentally sensitive fluids.

#190 · Material Handling and Process Systems

How do you adjust pump packing correctly?

Answer

Tighten the gland follower bolts evenly in small increments until leakage drops to 1 to 3 drops per second, which is the normal operating leak rate for lubrication and cooling. Over-tightening stops the drip but causes the packing to overheat, burn, and score the shaft sleeve. Check and adjust packing again after the first hour of operation on new packing.

#191 · Material Handling and Process Systems

What is a seal flush plan and why is it used?

Answer

A seal flush plan is a piping arrangement that circulates clean, cool liquid through the mechanical seal chamber to lubricate and cool the seal faces and to remove heat and debris. Without flushing, hot or dirty fluid can cause the seal faces to run dry, overheat, and fail quickly. The American Petroleum Institute (API) defines standard flush plans numbered for easy specification.

#192 · Material Handling and Process Systems

What does API flush Plan 11 do?

Answer

Plan 11 takes a small flow from the pump discharge, passes it through an orifice to control flow, and injects it into the seal chamber. This keeps the seal faces bathed in clean pump fluid at a pressure slightly above suction. It is the simplest and most common flush plan, suitable for clean, non-flashing liquids.

#193 · Material Handling and Process Systems

What does API flush Plan 21 do differently from Plan 11?

Answer

Plan 21 is similar to Plan 11 but adds a heat exchanger (cooler) in the flush line before the seal chamber. This cools the flush fluid before it contacts the seal faces, which is important when pumping hot liquids near their vapor pressure. Cooling the flush fluid lowers the vapor pressure risk and extends seal life.

#194 · Material Handling and Process Systems

What is API flush Plan 32 and when is it used?

Answer

Plan 32 injects clean fluid from an external source into the seal chamber instead of using the pump's own process fluid. It is used when the process fluid is dirty, abrasive, or hazardous and should not contact the seal faces. The external fluid must be compatible with the process and at a higher pressure than the seal chamber.

#195 · Material Handling and Process Systems

What is API flush Plan 53 and what type of seal does it support?

Answer

Plan 53 is a pressurized barrier fluid system used with a double mechanical seal. A clean, inert barrier fluid is maintained at a higher pressure than the process fluid between the inboard and outboard seal faces. If either seal leaks, the barrier fluid enters the process rather than the process fluid escaping to atmosphere, which is important for toxic or hazardous service.

#196 · Material Handling and Process Systems

What is a positive displacement pump and how does it differ from a centrifugal pump?

Answer

A positive displacement pump moves a fixed volume of liquid for every revolution or stroke, regardless of the pressure on the outlet. A centrifugal pump moves more liquid when outlet pressure is low and less when pressure is high. Positive displacement pumps are used when a constant flow rate is needed regardless of system pressure changes.

#197 · Material Handling and Process Systems

How does a gear pump work?

Answer

A gear pump uses two meshing gears inside a close-fitting housing. Liquid is carried in the spaces between the gear teeth around the outside from inlet to outlet. Where the teeth mesh, the liquid is squeezed out under pressure. Gear pumps produce steady flow and are used for lube oil, hydraulic fluid, and thick liquids.

#198 · Material Handling and Process Systems

How does a vane pump work?

Answer

A vane pump has a rotor with sliding vanes mounted off-center inside a circular housing. As the rotor turns, centrifugal force pushes the vanes outward against the housing wall, creating expanding and contracting chambers that trap and move liquid. Vane pumps are smooth-running and self-compensating for wear on the vane tips.

#199 · Material Handling and Process Systems

How does a piston pump differ from gear and vane pumps?

Answer

A piston pump uses one or more reciprocating pistons to draw liquid in on the backstroke and push it out on the forward stroke through check valves. Piston pumps can develop very high pressures and are used in high-pressure hydraulic systems and chemical injection applications. They require check valves to work, which must be kept clean and in good condition.

#200 · Material Handling and Process Systems

What is a lobe pump and what applications suit it?

Answer

A lobe pump uses two or more lobed rotors that mesh without touching to move liquid gently around the outside of the lobes from inlet to outlet. Because the rotors never contact each other, a lobe pump can handle soft solids, shear-sensitive products, and sterile fluids without damaging them. It is common in food processing, pharmaceuticals, and wastewater.

#201 · Material Handling and Process Systems

What are the main types of industrial compressors?

Answer

The four main types are reciprocating (piston), which compresses gas in a cylinder; rotary screw, which uses two meshing helical rotors; centrifugal, which uses a high-speed impeller to add velocity to the gas; and axial, which uses rows of blades to accelerate and diffuse the gas through the machine. Each type suits different pressure and flow requirements.

#202 · Material Handling and Process Systems

How does a rotary screw compressor work?

Answer

A rotary screw compressor has two helical rotors, a male and a female, that mesh and spin together inside a close-fitting housing. As the rotors turn, gas is trapped in the spaces between the threads and the housing and is compressed as the thread volume decreases toward the outlet. Oil-injected screws use oil to seal clearances and cool the gas during compression.

#203 · Material Handling and Process Systems

What is compressor valve maintenance on a reciprocating compressor?

Answer

Compressor valves are spring-loaded flap or plate valves that control gas flow into and out of each cylinder. They are the most commonly replaced wear item because the plates flex millions of times and fatigue. Inspect valves at scheduled intervals for cracked or broken plates, worn seats, and weakened springs; a leaking valve reduces compression efficiency and can cause overheating.

#204 · Material Handling and Process Systems

What is the purpose of a compressor intercooler?

Answer

An intercooler is a heat exchanger placed between stages of a multi-stage compressor to cool the gas back to near ambient temperature before it enters the next stage. Cooling the gas reduces its volume so the next stage does less work, improving overall efficiency. Intercoolers also remove moisture that condenses out of the gas as it cools, which must be drained regularly.

#205 · Material Handling and Process Systems

Why must intercooler moisture traps be drained regularly?

Answer

As hot compressed gas cools in the intercooler, water vapor in the gas condenses into liquid water that collects in the drain pot. If this water is not drained, it can carry over into the next compression stage and cause liquid slugging, which can break valves or damage the cylinder. Automatic drain traps are common but must be tested to confirm they are operating.

#206 · Material Handling and Process Systems

What is a belt conveyor and what are its main components?

Answer

A belt conveyor uses a continuous loop of flat or troughed belt running over a series of rollers (idlers) supported by a steel frame to carry bulk material or packages from one point to another. The main components are the head pulley (drive end), the tail pulley (return end), the idler rollers, the take-up device for tensioning the belt, and the drive unit.

#207 · Material Handling and Process Systems

What is the function of the head pulley on a belt conveyor?

Answer

The head pulley is the drive pulley at the discharge end of the conveyor. It is connected to the motor and gearbox and provides the traction that moves the belt. Material rides to the head pulley and discharges over it. The head pulley often has a rubber lagging surface to increase grip between the pulley and the belt.

#208 · Material Handling and Process Systems

What is a conveyor take-up device and how does it work?

Answer

A take-up device applies tension to the belt to prevent slippage on the drive pulley and to take up slack that develops as the belt stretches over time. Gravity take-ups use a weighted carriage that moves on rails to maintain constant tension automatically. Screw take-ups use threaded rods that must be adjusted manually when the belt stretches.

#209 · Material Handling and Process Systems

What causes a conveyor belt to track to one side?

Answer

A belt drifts to one side when the tension across its width is uneven or when idler rollers or pulleys are not square to the belt's travel direction. Common causes are a misaligned idler, a loaded idler that is seized and not turning, uneven loading onto the belt, and a take-up that is not centered. Correct by adjusting the training idlers, which tilt slightly to steer the belt back to center.

#210 · Material Handling and Process Systems

What is a conveyor belt splice and what types are used?

Answer

A belt splice joins the two ends of a conveyor belt to form a continuous loop. Mechanical fasteners use metal clips or staples driven through both belt ends for quick, easy splicing without heat. Vulcanized splices use heat and pressure to bond the rubber ends together, giving a stronger and more flexible joint that is preferred for long, heavily loaded conveyors.

#211 · Material Handling and Process Systems

How do you inspect a conveyor belt splice for integrity?

Answer

Look for separation of the splice edges, cracking, missing fastener clips, or rubber delamination at the joint. Run the belt slowly and listen for a thumping sound as the splice passes over each idler, which indicates it is not flat or is beginning to lift. A failing splice must be repaired before it opens completely, which would require an emergency shutdown.

#212 · Material Handling and Process Systems

What is a screw conveyor and what materials does it handle?

Answer

A screw conveyor moves bulk material along a trough using a rotating helical screw (auger) inside the trough. It is simple, compact, and well-suited for dry powders, granules, and semi-solid materials such as grain, cement, and wood chips. It is not suitable for fragile materials that would be damaged by the screw's mixing and tumbling action.

#213 · Material Handling and Process Systems

What is a hanger bearing on a screw conveyor?

Answer

A hanger bearing is an intermediate support bearing mounted from the trough cover that supports the screw shaft at intervals along long conveyors to prevent the screw from deflecting under its own weight. Because hanger bearings are submerged in the material being conveyed, they must be made from materials compatible with the product and often use self-lubricating bushings since re-greasing is impractical.

#214 · Material Handling and Process Systems

What is a bucket elevator and what is it used for?

Answer

A bucket elevator carries bulk material vertically using buckets attached to a belt or chain that runs between a head pulley at the top and a boot pulley at the bottom. Material is scooped up in the boot section and discharged at the head by centrifugal force. It is used to elevate grain, fertilizer, cement, and similar dry bulk materials.

#215 · Material Handling and Process Systems

What maintenance does the boot section of a bucket elevator require?

Answer

The boot section at the bottom of the elevator tends to accumulate fine material that the buckets do not pick up cleanly. This buildup must be cleaned out regularly to prevent material from packing under the belt or chain, which can cause belt damage, mistracking, and blockage. Inspect the boot pulley bearings, the take-up adjustment, and the bucket attachment bolts during each maintenance visit.

#216 · Material Handling and Process Systems

What is a drag chain conveyor and how does it differ from a belt conveyor?

Answer

A drag chain conveyor uses one or two strands of heavy chain with attached flights or paddles to drag bulk material along the bottom of an enclosed trough. Unlike a belt conveyor, there is no belt surface; the material sits directly in the trough. Drag chain conveyors handle hot, abrasive, or heavy materials that would damage a rubber belt.

#217 · Material Handling and Process Systems

What is a P&ID; and what does it show?

Answer

A P&ID; stands for Piping and Instrumentation Diagram. It is a detailed drawing of a process system showing all pipes, equipment, valves, instruments, and control loops using standardized symbols. A millwright uses a P&ID; to understand how the system is connected, find isolation valves, and trace the flow path before working on any component.

#218 · Material Handling and Process Systems

What does a gate valve do and how is it different from a globe valve?

Answer

A gate valve uses a flat or wedge-shaped disc that slides perpendicular to the flow to open or block the pipe fully. It is used for isolation only; it is not suitable for throttling. A globe valve has a plug that moves parallel to the flow and is designed to regulate flow rate. Globe valves have higher pressure drop but better throttling control than gate valves.

#219 · Material Handling and Process Systems

What is a ball valve and what are its advantages?

Answer

A ball valve uses a drilled sphere that rotates 90 degrees to align its bore with the pipe for full open or turn it perpendicular for full closed. It opens and closes quickly, has very low pressure drop when fully open, and seals tightly. Ball valves are the most common isolation valve in modern industrial piping.

#220 · Material Handling and Process Systems

What is a butterfly valve and where is it used?

Answer

A butterfly valve uses a disc mounted on a shaft in the center of the pipe bore that rotates 90 degrees to open or close. It is compact and lightweight compared to a gate or ball valve of the same size. Butterfly valves are used for large pipe sizes, low-pressure systems, and flow throttling applications such as air handling and water treatment.

#221 · Material Handling and Process Systems

What is a check valve and when does it open or close?

Answer

A check valve allows flow in only one direction and closes automatically when flow tries to reverse. Common types include swing check, where a hinged disc swings open with forward flow and gravity-closes when flow stops, and lift check, where a piston lifts off its seat with forward flow. Check valves protect pumps from reverse flow when the pump stops.

#222 · Material Handling and Process Systems

What is a needle valve and where is it used?

Answer

A needle valve uses a tapered pin (needle) that threads into a seat to give very fine, precise control over a small flow rate. It is used in instrument tubing, sampling lines, and calibration connections where the flow must be adjusted accurately. Needle valves are not suitable for large flow or high-velocity applications because the tapered seat is easily eroded.

#223 · Material Handling and Process Systems

What is valve packing and how does it seal the stem?

Answer

Valve packing is a deformable material, typically graphite, PTFE, or braided fiber, compressed into the stuffing box around the valve stem by a gland follower. The compression forces the packing material to conform to the stem surface and the stuffing box wall, blocking process fluid from escaping along the stem. Packing must be tightened or replaced when leakage appears around the stem.

#224 · Material Handling and Process Systems

What is a pressure relief valve (PRV) and why is it required?

Answer

A PRV is a safety device that automatically opens to release pressure when the system exceeds a preset limit, preventing vessel or piping rupture. It reseats and closes when pressure drops back below the set point. PRVs are required by pressure vessel codes on any closed system that contains a fluid that could be heated or pressurized beyond the vessel's design rating.

#225 · Material Handling and Process Systems

How is a PRV set point tested in the field?

Answer

A PRV can be tested using a hand pump and pressure gauge connected to a test stand to verify the valve opens at its stamped set pressure. Some valves have a test lever that lifts the disc manually to confirm the valve is not stuck. PRVs should be tested and recertified on a scheduled interval as required by local pressure vessel regulations.

#226 · Material Handling and Process Systems

What is an expansion joint in a piping system?

Answer

An expansion joint is a flexible section, often made from bellows, rubber, or fabric, installed in a pipe run to absorb thermal expansion and contraction as the pipe heats and cools. Without expansion joints, long straight pipe runs would build up enormous stresses as the metal grows. Expansion joints also absorb vibration and reduce forces on connected equipment.

#227 · Material Handling and Process Systems

What types of expansion joints are used in industrial piping?

Answer

Metal bellows expansion joints handle high temperatures and pressures and are used in steam, hot oil, and gas service. Rubber expansion joints absorb vibration and handle lower temperatures and pressures, common on pump connections. Fabric expansion joints are used on large-diameter low-pressure ductwork such as combustion air and flue gas systems.

#228 · Material Handling and Process Systems

What is a flanged pipe connection and what does the bolt torque sequence involve?

Answer

A flanged connection uses two matching ring faces with a gasket between them, bolted together to make a pressure-tight joint. The bolts must be tightened in a cross or star pattern, not around the flange in order, to load the gasket evenly and prevent it from being compressed more on one side than the other. Multiple passes at increasing torque values are used to achieve uniform gasket compression.

#229 · Material Handling and Process Systems

What gasket materials are used for different pipe services?

Answer

Spiral wound metallic gaskets are used for high-pressure, high-temperature steam and process piping. Full-face rubber or elastomeric gaskets are used for water, low-pressure air, and mild chemicals. PTFE envelope gaskets are used for corrosive chemical service. The gasket material must be compatible with the fluid and rated for the temperature and pressure of the service.

#230 · Material Handling and Process Systems

What is carbon steel pipe and what are its limitations?

Answer

Carbon steel pipe is the most common pipe material in industrial plants because it is strong, affordable, and weld-able. Its main limitation is corrosion; it will rust rapidly if exposed to water, acids, or certain chemicals without a protective coating or lining. Carbon steel is also not suitable for very low temperatures because it can become brittle.

#231 · Material Handling and Process Systems

When is stainless steel pipe selected over carbon steel?

Answer

Stainless steel is selected when corrosion resistance is required, such as in food and beverage, pharmaceutical, and chemical service with acids or caustics that would corrode carbon steel. It maintains its strength at both high and low temperatures better than carbon steel. The main trade-off is cost; stainless steel can be three to five times more expensive per metre than carbon steel.

#232 · Material Handling and Process Systems

What is HDPE pipe and what are its advantages in process piping?

Answer

HDPE stands for High Density Polyethylene. It is a plastic pipe that resists corrosion from a wide range of chemicals, is lightweight, and is joined by heat fusion rather than welding or flanges. It is used for water treatment, chemical drainage, and underground service. Its limitation is that it cannot handle high temperatures; most HDPE pipe is limited to 60 degrees Celsius.

#233 · Material Handling and Process Systems

What is a fixed pipe support versus a guide support?

Answer

A fixed support, also called an anchor, is attached rigidly to the pipe and the structure so the pipe cannot move in any direction at that point. A guide support holds the pipe in place laterally but allows it to move axially along its length so thermal expansion can occur freely. Both types are shown on the piping isometric drawing and must be installed at the specified locations.

#234 · Material Handling and Process Systems

What is a spring hanger support and when is it needed?

Answer

A spring hanger uses a coil spring or constant-support spring assembly to support the pipe while allowing it to move vertically as it expands and contracts thermally. It is needed on hot piping systems, such as steam lines, where the pipe rises significantly when heated. Without spring hangers, a hot pipe that is forced into a rigid support would create damaging stresses in the pipe and connected equipment.

#235 · Material Handling and Process Systems

What is a shell-and-tube heat exchanger and how does it work?

Answer

A shell-and-tube heat exchanger has a bundle of tubes mounted inside a cylindrical shell. One fluid flows inside the tubes and another flows over the outside of the tubes within the shell. Heat transfers through the tube walls from the hotter fluid to the cooler fluid. The two fluids can flow in the same direction (parallel flow) or opposite directions (counterflow), with counterflow giving better heat transfer.

#236 · Material Handling and Process Systems

How do you inspect a shell-and-tube heat exchanger for tube leaks?

Answer

A hydrostatic test pressurizes one side (usually the tube side) with water while the other side is open. Any wet spots, drips, or pressure drop indicate a leaking tube. Individual leaking tubes can be plugged with tapered metal plugs to temporarily return the exchanger to service, but a tube bundle with more than 10 percent of tubes plugged is normally scheduled for retubing.

#237 · Material Handling and Process Systems

What is a plate-and-frame heat exchanger and what are its advantages?

Answer

A plate-and-frame heat exchanger uses a stack of corrugated metal plates clamped together in a frame, with alternating channels carrying the hot and cold fluids. The large surface area of the plates makes them very compact and efficient compared to shell-and-tube exchangers. They are also easy to clean and expand by adding more plates, but are limited to lower pressures and temperatures.

#238 · Material Handling and Process Systems

What is fouling in a heat exchanger and how does it reduce performance?

Answer

Fouling is the buildup of scale, biological growth, corrosion products, or process deposits on the heat transfer surfaces inside a heat exchanger. This layer acts as insulation and reduces the rate of heat transfer, meaning the exchanger cannot cool or heat the process fluid to the required temperature. Fouling also increases pressure drop through the exchanger and can accelerate corrosion under the deposit.

#239 · Material Handling and Process Systems

How is heat exchanger fouling cleaned?

Answer

Mechanical cleaning uses high-pressure water jetting, brushes, or scrapers to remove scale and deposits from tube surfaces. Chemical cleaning circulates an acid or alkaline solution through the exchanger to dissolve deposits that are too hard for mechanical methods. The choice of method depends on the type of fouling; calcium scale dissolves well in dilute acid, while biological growth responds to alkaline cleaners or biocides.

#240 · Material Handling and Process Systems

What is the purpose of baffles inside a shell-and-tube heat exchanger?

Answer

Baffles are plates inside the shell that direct the shell-side fluid to flow across the tubes rather than straight through the shell from end to end. Crossflow over the tubes gives much better heat transfer than straight flow along the tubes. Baffles also support the tube bundle and prevent vibration damage to the tubes from shell-side flow.

#241 · Material Handling and Process Systems

What is pump recirculation and why can it damage the pump?

Answer

Recirculation happens when a centrifugal pump runs well below its minimum flow, causing flow reversal and turbulence at the impeller eye or discharge vanes. This turbulent flow creates high-energy cavitation-like erosion and high radial shaft loads that damage the impeller, seal, and bearings. A minimum flow bypass line or a recirculation control valve is used to prevent the pump from running below its minimum safe flow.

#242 · Material Handling and Process Systems

What is a centrifugal pump operating in parallel and what happens to the flow?

Answer

When two identical pumps operate in parallel, both pump into the same discharge header. The total flow rate increases but the head (pressure) stays roughly the same as one pump alone. Parallel pumping is used to increase flow capacity or provide standby redundancy. Both pumps must have similar curves or one pump may be pushed to a low-flow condition that causes damage.

#243 · Material Handling and Process Systems

What is a centrifugal pump operating in series and what happens to the head?

Answer

When two pumps operate in series, the discharge of the first pump feeds the inlet of the second. The second pump adds more pressure on top of what the first already developed, so the total head (pressure) roughly doubles while flow stays the same. Series arrangements are used when the required pressure is higher than one pump can produce on its own.

#244 · Material Handling and Process Systems

What is a flow meter and what types are common in industrial piping?

Answer

A flow meter measures the rate at which fluid passes through a pipe. Common types include the orifice plate, which measures pressure drop across a restriction; the magnetic flow meter (magmeter), which measures the voltage generated by a conductive liquid moving through a magnetic field; and the turbine meter, which counts blade rotations to calculate flow. The right type depends on the fluid properties and required accuracy.

#245 · Material Handling and Process Systems

What is a pressure transmitter and how is it connected to a pipe?

Answer

A pressure transmitter is an instrument that senses the pressure in a pipe or vessel and sends an electrical signal (usually 4-20 mA) to a control system. It connects to the pipe through an instrument tap and isolation valve. The tap location matters; pressure taps on liquid lines should be on the side of the pipe, not the top (where gas can collect) or the bottom (where solids settle).

#246 · Material Handling and Process Systems

What is a solenoid valve and how does it work?

Answer

A solenoid valve uses an electromagnetic coil to pull a metal plunger that opens or closes a valve seat when electrical current is applied. When the current is removed, a spring returns the plunger to its original position. Solenoid valves are used to automate the opening and closing of small lines quickly from a remote control system.

#247 · Material Handling and Process Systems

What is a control valve and how does it differ from an on-off valve?

Answer

A control valve can be positioned anywhere between fully open and fully closed to regulate flow at any desired rate. It is driven by an actuator (pneumatic, electric, or hydraulic) responding to a signal from a controller. An on-off valve is either fully open or fully closed with nothing in between. Control valves are used in closed-loop process control; on-off valves are used for isolation.

#248 · Material Handling and Process Systems

What is pipe stress and why must it be managed?

Answer

Pipe stress is the force, bending moment, or torsion that acts on a pipe due to thermal expansion, weight, pressure, and external loads. If pipe stress is too high, welds crack, flanges leak, and connected equipment like pumps and compressors is damaged by the forces transmitted through the nozzle connections. Pipe supports, expansion joints, and expansion loops are designed to keep stress within allowable limits.

#249 · Material Handling and Process Systems

What is an air receiver tank and what maintenance does it require?

Answer

An air receiver tank stores compressed air from a compressor and smooths out pressure pulsations before the air is distributed to the plant. It must be drained regularly to remove condensed water that collects at the bottom, which causes rust and corrosion from the inside. The tank is also a pressure vessel that must be inspected and certified on a schedule set by the provincial pressure vessel authority.

#250 · Material Handling and Process Systems

What is water hammer in a piping system and how is it prevented?

Answer

Water hammer is the pressure shock wave created when a flowing liquid is stopped suddenly, such as when a valve closes too fast. The moving column of liquid has momentum, and when it stops abruptly the kinetic energy converts to a pressure spike that can crack pipes, damage fittings, and destroy valve seats. It is prevented by using slow-closing valves, installing surge relief valves, and keeping flow velocities within recommended limits.

#251 · Material Handling and Process Systems

What type of seal is used on a tank agitator shaft?

Answer

Tank agitators use mechanical seals or packed gland seals where the shaft enters the tank. Mechanical seals are preferred for hazardous or high-pressure fluids because they reduce leakage. A lip seal is not suitable for agitator shafts due to the side loads from the impeller.

#252 · Material Handling and Process Systems

How do you inspect and replace packing on an agitator stuffing box?

Answer

Remove the gland follower and pull out the old packing rings using a packing hook. Clean the stuffing box bore and shaft before installing new rings, staggering the cuts 90 degrees apart. Tighten the gland evenly and allow a small controlled drip to lubricate the packing.

#253 · Material Handling and Process Systems

What is a pitched-blade impeller and when is it used?

Answer

A pitched-blade impeller has blades angled at roughly 45 degrees to the shaft axis. It creates both axial (up-down) and radial (outward) flow, making it good for blending and suspending solids. It is a common choice for general mixing in tanks.

#254 · Material Handling and Process Systems

What is a Rushton turbine impeller and what flow pattern does it produce?

Answer

A Rushton turbine has flat vertical blades mounted on a disc, producing strong radial flow outward from the impeller. It is widely used for gas dispersion and chemical reactions where high shear is needed. The disc prevents gas bubbles from short-circuiting up the shaft.

#255 · Material Handling and Process Systems

Why is a pump baseplate grouted after installation?

Answer

Grouting fills the gap between the baseplate and the foundation, transferring vibration loads evenly into the concrete. It prevents the baseplate from flexing or shifting under operating conditions. Epoxy grout is the preferred type for precision machinery because it does not shrink as it cures.

#256 · Material Handling and Process Systems

What is the correct procedure for grouting a pump baseplate?

Answer

Level and shim the baseplate first, then form a dam around it and pour non-shrink epoxy grout to the underside. Allow the grout to fully cure before tightening anchor bolts to final torque. Check alignment again after grouting because curing can shift the baseplate slightly.

#257 · Material Handling and Process Systems

What is soft foot and what problems does it cause on a pump?

Answer

Soft foot means one or more of a machine's feet do not make full contact with the baseplate when the hold-down bolts are loose. When you tighten the bolts, the frame twists, causing misalignment and bearing overload. It must be corrected with shims before performing shaft alignment.

#258 · Material Handling and Process Systems

How do you check for and correct soft foot on a pump?

Answer

Place a dial indicator on the shaft or bearing housing, then loosen each foot bolt one at a time and record the shaft movement. Any movement greater than 0.05 mm (0.002 in) at that foot indicates soft foot. Correct it by adding precision shims under the offending foot until all readings are within tolerance.

#259 · Material Handling and Process Systems

What are the main differences between laser and dial indicator shaft alignment?

Answer

Dial indicator alignment uses mechanical gauges and requires manual calculations to find the correction moves. Laser alignment uses beam detectors that feed data to a computer, which calculates corrections automatically and in real time. Laser systems are faster, reduce math errors, and give live readout as you move the machine.

#260 · Material Handling and Process Systems

What is the purpose of a coupling spacer on a pump and how is it removed?

Answer

A coupling spacer is a removable center section that creates a gap between the pump and driver hubs, allowing the mechanical seal or packing to be removed without moving either machine. To remove it, loosen the set screws or bolts on both hubs, then slide the spacer out of the gap. Always lock out the driver before any coupling work.

#261 · Material Handling and Process Systems

What is the correct start-up sequence for a centrifugal pump?

Answer

Prime the pump and open the suction valve fully before starting. Start the motor with the discharge valve partially closed, then slowly open the discharge valve to the desired flow rate. Never run a centrifugal pump against a closed discharge for more than a few seconds or it will overheat.

#262 · Material Handling and Process Systems

How do you perform a pump performance test using head, flow, and power?

Answer

Measure suction and discharge pressures, then calculate total head in metres. Measure flow rate using a flow meter or timed fill method. Calculate hydraulic power (kW) as flow times head times specific gravity divided by a constant, then compare to the pump curve to confirm the pump is operating at the correct duty point.

#263 · Material Handling and Process Systems

What is the difference between a strainer mesh size and a filter micron rating?

Answer

A strainer mesh size refers to the number of wires per inch; a higher mesh number means smaller openings and finer filtration. A filter micron rating directly states the size of particles that are captured, where a lower micron number captures finer particles. Strainers handle coarser protection (e.g., 40 mesh) while filters provide fine clean-up downstream.

#264 · Material Handling and Process Systems

How do you select the correct micron rating for a process filter?

Answer

The micron rating must be finer than the particle size that would damage downstream equipment such as seals, valves, or bearings. Check the equipment manufacturer's specification for the required cleanliness level. A balance must be struck between fine filtration and acceptable pressure drop across the filter element.

#265 · Material Handling and Process Systems

How do you inspect a shell-and-tube heat exchanger for tube leaks?

Answer

Isolate and drain the heat exchanger, then pressurize one side with air or nitrogen and apply soapy water to the tube sheet to look for bubbles. Alternatively, use a hydrostatic test by filling one side with water under pressure and checking the other side for water ingress. Failed tubes are plugged with approved tube plugs to restore service.

#266 · Material Handling and Process Systems

When is a heat exchanger tube plugged instead of replaced?

Answer

A tube is plugged when it has a single leak and replacement of the full bundle is not practical or cost-effective. Most manufacturers allow up to 10 percent of tubes to be plugged before the exchanger must be re-tubed. Plugging reduces heat transfer area, so the impact on process performance must be evaluated.

#267 · Material Handling and Process Systems

What causes cavitation in a centrifugal pump and how is it identified?

Answer

Cavitation occurs when the pressure at the pump inlet drops below the vapour pressure of the liquid, causing bubbles to form and then collapse violently inside the pump. It sounds like gravel is being pumped and causes rapid erosion of the impeller and casing. The fix is to increase suction pressure, reduce flow restriction on the suction side, or lower the pump speed.

#268 · Material Handling and Process Systems

What is NPSH and why does it matter when installing a pump?

Answer

NPSH stands for Net Positive Suction Head, which is the pressure available at the pump inlet above the vapour pressure of the liquid. The available NPSH (NPSHa) from the system must always exceed the required NPSH (NPSHr) stated on the pump curve. If NPSHa is too low, the pump will cavitate and be damaged.

#269 · Material Handling and Process Systems

What maintenance checks are performed on a tank agitator drive gearbox?

Answer

Check the oil level and condition in the gearbox regularly, looking for discolouration or a burnt smell that indicates overheating or contamination. Inspect the output shaft seal for leaks and listen for unusual noise during operation. Change the gearbox oil at the manufacturer's recommended interval or after water contamination is found.

#270 · Material Handling and Process Systems

How do you align a top-entry agitator shaft during installation?

Answer

The agitator shaft must be perpendicular to the tank flange face to prevent side loading on the seal and bearings. Use a precision level or dial indicator on the shaft while adjusting the drive mounting. Check runout at the lower end of the shaft to confirm it is within the manufacturer's specification before sealing.

#271 · Fluid Power Systems

State Pascal's Law and explain what it means in a hydraulic system.

Answer

Pascal's Law states that pressure applied to a confined fluid is transmitted equally in all directions throughout the fluid. In a hydraulic system this means the pressure generated by the pump acts on every surface in the circuit simultaneously. This is what allows a small cylinder to lift a heavy load when its area is large.

#272 · Fluid Power Systems

What are the three main categories of hydraulic fluid and when is each used?

Answer

Mineral oil is the most common type, offering good lubrication and low cost for standard industrial machines. Fire-resistant fluids (water-glycol, phosphate ester, or water-in-oil emulsion) are used near high heat sources or open flames. Biodegradable fluids (vegetable-based or synthetic ester) are used in environmentally sensitive locations such as forestry or near waterways.

#273 · Fluid Power Systems

What is the ISO 4406 cleanliness code and how do you read it?

Answer

ISO 4406 rates hydraulic fluid cleanliness using three numbers separated by slashes, such as 18/16/13. Each number represents the count of particles per millilitre at 4 micron, 6 micron, and 14 micron sizes respectively, expressed as a range code. Lower numbers mean cleaner fluid; most hydraulic servo systems require 16/14/11 or cleaner.

#274 · Fluid Power Systems

How does a hydraulic gear pump produce flow?

Answer

Two meshing gears rotate in opposite directions inside a close-tolerance housing. Fluid is trapped in the spaces between the gear teeth and the housing on the inlet side, then carried around to the outlet where the teeth mesh together and push the fluid out. Gear pumps are fixed displacement, simple, and tolerant of contamination.

#275 · Fluid Power Systems

How does a vane pump differ from a gear pump in its operating principle?

Answer

A vane pump uses a slotted rotor with spring- or pressure-loaded vanes that slide in and out as the rotor turns inside an elliptical cam ring. The changing volume between vanes creates suction on one side and displacement on the other. Vane pumps run more quietly than gear pumps and can be balanced for lower bearing loads.

#276 · Fluid Power Systems

How does a fixed-displacement axial piston pump work?

Answer

An axial piston pump uses pistons arranged parallel to the drive shaft, reciprocated by a fixed-angle swash plate as the cylinder barrel rotates. Each piston draws fluid in through a port plate kidney slot on the suction stroke and pushes it out on the pressure stroke. Displacement is set by the swash plate angle and cannot be changed.

#277 · Fluid Power Systems

What makes a variable-displacement piston pump different from a fixed one?

Answer

A variable-displacement pump has an adjustable swash plate angle that changes the stroke of each piston, allowing the output flow to be varied from zero to maximum without changing pump speed. This is controlled by a pressure-compensator, load-sensing signal, or manual control. Variable pumps save energy because they only deliver the flow the system demands.

#278 · Fluid Power Systems

How do you calculate theoretical pump flow from displacement and speed?

Answer

Multiply the pump's displacement per revolution (in litres or cubic inches) by its shaft speed in RPM to get theoretical flow. For example, a 28 cc/rev pump running at 1500 RPM delivers 42 L/min theoretically. Actual flow is lower due to internal leakage, expressed as volumetric efficiency.

#279 · Fluid Power Systems

What is the purpose of a hydraulic relief valve and where is it located?

Answer

A relief valve limits maximum system pressure by opening to dump excess flow back to the tank when pressure reaches the set point. It is always located between the pump outlet and the first directional valve, acting as the system's safety device. The relief valve setting must be higher than the maximum working pressure but below the pressure rating of the weakest component.

#280 · Fluid Power Systems

What is an unloading valve and when is it used?

Answer

An unloading valve dumps pump flow back to tank at near-zero pressure when a signal (usually from an accumulator reaching full charge) tells the pump it is not needed. This saves energy and reduces heat build-up by not forcing the pump to work against the relief valve continuously. It is common in accumulator circuits and hi-lo pump circuits.

#281 · Fluid Power Systems

What does a sequence valve do in a hydraulic circuit?

Answer

A sequence valve ensures one actuator completes its stroke before the next one begins, by holding back flow to the second circuit until the first circuit reaches the valve's set pressure. It has an external drain because its outlet is not at tank pressure. Sequence valves are used in clamping and drilling circuits where order of operations matters.

#282 · Fluid Power Systems

What is a counterbalance valve and what load condition requires it?

Answer

A counterbalance valve holds a load in position by blocking cylinder return flow until enough pilot pressure is applied from the opposite port to open it. It is required on vertically mounted cylinders supporting gravity loads to prevent the load from free-falling if a hose bursts. The counterbalance setting is typically 1.3 times the maximum load-induced pressure.

#283 · Fluid Power Systems

What is a pressure-reducing valve and how does it differ from a relief valve?

Answer

A pressure-reducing valve limits the pressure in one branch of a circuit to a lower value than the main system pressure. Unlike a relief valve, it is normally open and closes partially as downstream pressure reaches the set point, while a relief valve is normally closed and opens when pressure is too high. Pressure-reducing valves are used to protect sensitive downstream components.

#284 · Fluid Power Systems

What does the designation 4/3 mean when describing a directional control valve?

Answer

The first number (4) is the number of ports on the valve (P, T, A, B). The second number (3) is the number of switching positions (left, centre, right). A 4/3 valve is the most common type used to control a double-acting cylinder, connecting pressure and tank to either side of the cylinder or holding the cylinder in a centre condition.

#285 · Fluid Power Systems

What are the three common 4/3 valve centre conditions and when is each chosen?

Answer

An open centre connects all four ports together at mid-position, allowing the pump to unload to tank at low pressure. A closed centre blocks all four ports, holding cylinder position and keeping the pump at full pressure. A tandem centre connects P to T (unloading the pump) while blocking both actuator ports to hold the load.

#286 · Fluid Power Systems

How does a solenoid actuator shift a directional control valve?

Answer

An electric solenoid coil creates a magnetic field that pulls an iron plunger, which pushes the valve spool to the shifted position. When the solenoid is de-energised, a return spring pushes the spool back to its neutral position. Solenoid valves are controlled by PLCs or electrical switches and allow remote and automated operation.

#287 · Fluid Power Systems

What is a pilot-operated directional control valve and why is it used?

Answer

A pilot-operated valve uses hydraulic pressure from a small pilot signal to shift the main spool, rather than a direct solenoid. This allows a small solenoid to control a large high-flow valve without needing a heavy solenoid. Pilot-operated valves are used in high-flow circuits where a direct-acting solenoid would be too large.

#288 · Fluid Power Systems

What is a flow control valve and how does it regulate cylinder speed?

Answer

A flow control valve restricts the flow of oil to or from a hydraulic actuator by creating a variable orifice, which limits how fast the cylinder can move. Reducing the orifice slows the actuator; opening it allows more flow and higher speed. Flow control valves can be placed in the meter-in or meter-out position depending on load conditions.

#289 · Fluid Power Systems

What is a pressure-compensated flow control valve?

Answer

A pressure-compensated flow control maintains a constant flow rate regardless of changes in inlet or load pressure. It contains a pressure-compensating spool that automatically adjusts the orifice opening to keep the pressure drop across the throttle constant. This results in consistent actuator speed even when load changes during the stroke.

#290 · Fluid Power Systems

What is a deceleration valve and where is it used in a circuit?

Answer

A deceleration valve is a normally-open flow control valve with a cam-operated actuator that gradually closes as a machine part approaches the end of its stroke, slowing the actuator smoothly. It is mounted on the machine frame so the moving part's cam depresses it at the right position. This reduces shock and wear at end of stroke.

#291 · Fluid Power Systems

What is the function of a check valve in a hydraulic circuit?

Answer

A check valve allows flow in one direction only and blocks it completely in the reverse direction, acting like a one-way door for hydraulic oil. It is used to prevent reverse flow from a loaded actuator back to the pump, and to separate parallel pump circuits. Check valves have a cracking pressure (the minimum pressure needed to open them).

#292 · Fluid Power Systems

How does a pilot-operated check valve (POCV) differ from a standard check valve?

Answer

A POCV blocks reverse flow like a standard check valve, but it can be opened in reverse by applying a hydraulic pilot signal to an external pilot port. This allows the cylinder to retract on command even though the check valve would normally hold it. POCVs are used to lock cylinder position securely and release it only when commanded.

#293 · Fluid Power Systems

What is the difference between a single-acting and a double-acting hydraulic cylinder?

Answer

A single-acting cylinder has hydraulic pressure on one side only; it extends under pressure and retracts by gravity, a spring, or an external load. A double-acting cylinder has hydraulic connections on both sides so it can be powered in both the extend and retract directions. Double-acting cylinders are used where force is needed in both directions.

#294 · Fluid Power Systems

How does a telescoping cylinder work and where is it used?

Answer

A telescoping cylinder has two or more nested stages that extend one at a time, allowing a very long stroke from a compact retracted length. The largest stage extends first because it has the greatest area. They are common on dump truck bodies and lifting platforms where installation space is limited.

#295 · Fluid Power Systems

What is cylinder cushioning and why is it built into hydraulic cylinders?

Answer

Cushioning is a built-in feature that traps a small volume of oil at the end of the stroke, forcing it through a small adjustable needle valve to slow the piston before it hits the end cap. This absorbs kinetic energy and prevents metal-to-metal impact that would damage the cylinder and machine. The cushion needle is adjusted to give smooth deceleration without rebound.

#296 · Fluid Power Systems

How do you calculate the extend force of a hydraulic cylinder?

Answer

Multiply the system pressure (in Pa) by the full bore area of the piston (in square metres) to get force in Newtons. For example, a 100 mm bore cylinder at 200 bar (20 MPa) has an area of 0.00785 square metres, producing about 157,000 N (157 kN). The retract force is less because the rod area reduces the effective piston area on the rod side.

#297 · Fluid Power Systems

How do you calculate the differential pressure area effect on a cylinder in regenerative mode?

Answer

In regenerative mode, the return oil from the rod side is fed back into the cap side instead of returning to tank. The effective area pushing the piston forward is only the rod cross-section area (cap area minus rod area). This doubles the speed but reduces force, which is useful for rapid advance before a work stroke.

#298 · Fluid Power Systems

What is the purpose of a hydraulic accumulator?

Answer

An accumulator stores hydraulic energy in the form of compressed gas (pre-charged with nitrogen) and releases it quickly when the circuit demands it. It is used to supplement pump flow for short peak demand cycles, maintain pressure during pump shutdown, and absorb pressure shocks (pulsation dampening). Never use oxygen or air to pre-charge an accumulator.

#299 · Fluid Power Systems

What are the three types of hydraulic accumulators and how do they differ?

Answer

A bladder accumulator has a rubber bladder separating the gas from the oil, giving fast response and a wide usable volume range. A piston accumulator uses a sliding piston to separate gas and oil, handling large volumes but responding more slowly. A diaphragm accumulator uses a moulded rubber diaphragm and is compact, used for low-volume pulsation dampening.

#300 · Fluid Power Systems

How do you safely pre-charge a bladder accumulator with nitrogen?

Answer

Fully discharge the hydraulic oil from the accumulator by relieving system pressure before touching the gas valve. Connect a nitrogen charging kit to the Schrader-type gas valve and set the pre-charge to the manufacturer's specification, typically 60 to 90 percent of the minimum system working pressure. Never use compressed air because oxygen combined with hydraulic oil under pressure creates an explosion hazard.

#301 · Fluid Power Systems

How does a hydraulic gear motor work and what are its limitations?

Answer

A hydraulic gear motor works like a gear pump in reverse: pressurised oil entering the housing pushes the gear teeth apart, causing the gears to rotate and turn the output shaft. Gear motors are simple, low-cost, and tolerant of contamination, but they are less efficient than piston motors and have fixed displacement so speed can only be changed by varying flow. They are used for low-torque, continuous-duty applications.

#302 · Fluid Power Systems

How do you calculate the output torque of a hydraulic motor?

Answer

Torque equals pressure (in bar) multiplied by displacement (in cc/rev) divided by 62.8. For example, a 50 cc/rev motor at 200 bar produces approximately 159 Nm of theoretical torque. Actual torque is lower due to mechanical efficiency losses, typically 90 to 95 percent for a piston motor.

#303 · Fluid Power Systems

What is the difference between an axial piston motor and a radial piston motor?

Answer

An axial piston motor has pistons arranged parallel to the shaft, reciprocated by a swash plate or bent axis; it runs at high speed and is efficient, suited for vehicle drives and winches. A radial piston motor has pistons arranged around the shaft like spokes pointing outward, giving very high torque at low speed without a gearbox. Radial piston motors are used for wheel motors and slow-speed heavy drives.

#304 · Fluid Power Systems

What do the ISO 1219 hydraulic circuit symbols represent for a pump?

Answer

A hydraulic pump is shown as a circle with a solid triangle pointing away from the circle toward the pressure line, indicating flow direction. A fixed-displacement pump has one triangle; a variable-displacement pump has an arrow drawn across the circle. The shaft drive line connects at the centre of the circle.

#305 · Fluid Power Systems

How is a hydraulic cylinder represented in an ISO 1219 circuit diagram?

Answer

A double-acting cylinder is drawn as a rectangle with a rod line extending from one end, with port connections shown at each end. The rod end is indicated by a shorter rectangle representing the rod attached to the piston. Single-acting cylinders show only one port connection and sometimes a spring symbol for the return.

#306 · Fluid Power Systems

What is the difference between an open-centre and a closed-centre hydraulic circuit?

Answer

In an open-centre circuit, pump flow passes through the directional valve and back to tank at low pressure when no work is being done, keeping the pump unloaded. In a closed-centre circuit, the directional valve blocks all ports in neutral so pump flow cannot return to tank, forcing the pump to work against full pressure unless an unloading valve is fitted. Open-centre suits fixed-displacement pumps; closed-centre suits variable-displacement pumps with pressure compensation.

#307 · Fluid Power Systems

How does a load-sensing hydraulic system improve efficiency?

Answer

A load-sensing system signals the pump's displacement control with the actual pressure required by the active actuator, so the pump only generates pressure slightly above the load requirement rather than maintaining full system pressure all the time. This reduces heat generation and power consumption significantly when loads are light or partial. The pump output closely tracks real demand.

#308 · Fluid Power Systems

What is a proportional valve and how does it differ from a regular solenoid valve?

Answer

A proportional valve uses a specially wound solenoid coil whose force is proportional to the electrical current sent to it, allowing the spool to be positioned anywhere between fully closed and fully open. A regular on/off solenoid valve only has two states: fully shifted or spring-centred. Proportional valves allow smooth variable control of flow and pressure from a PLC.

#309 · Fluid Power Systems

What is a servo valve and in what applications is it used?

Answer

A servo valve is a high-precision proportional valve that responds to a very small electrical input signal and uses a torque motor and hydraulic amplifier stage to position the spool with great accuracy and speed. Servo valves are used in closed-loop position control systems such as injection moulding machines, test equipment, and aircraft actuators. They require very clean hydraulic fluid (typically ISO 16/14/11 or cleaner).

#310 · Fluid Power Systems

Where are suction, pressure, and return filters located in a hydraulic system?

Answer

A suction strainer or filter is placed in the tank before the pump inlet to remove large particles that could damage the pump. A pressure filter is placed after the pump outlet to capture fine particles before they reach sensitive valves. A return filter is placed in the return line before the tank to clean oil before it re-enters the reservoir, and is the most common type.

#311 · Fluid Power Systems

How do you know when a hydraulic filter element needs to be changed?

Answer

Most hydraulic filters have a built-in bypass indicator, either a pop-up pin, a pressure gauge, or an electrical switch, that activates when the pressure drop across the element reaches a set limit (typically 3.5 to 7 bar). Change the element before the indicator activates if following a time-based schedule, or immediately after the indicator signals. Always change the element during a scheduled maintenance window, not just when it bypasses.

#312 · Fluid Power Systems

What is the function of a heat exchanger in a hydraulic system?

Answer

A hydraulic heat exchanger (oil cooler) removes heat generated by inefficiencies in the system, such as pressure drops across valves and pump losses, to keep the oil temperature in the ideal operating range (typically 40 to 60 degrees C). Overheated oil loses viscosity, breaks down chemically, and causes accelerated wear. Coolers use either water or air to remove heat from the oil.

#313 · Fluid Power Systems

What is the maximum allowable bend radius rule for hydraulic hose?

Answer

Every hydraulic hose has a minimum bend radius specified by the manufacturer; bending tighter than this crushes the reinforcement layers and causes premature failure. As a general rule, the minimum bend radius is typically six to twelve times the hose's outside diameter depending on the hose construction and pressure rating. Hose must also not be twisted or kinked during installation.

#314 · Fluid Power Systems

What are the common types of hydraulic hose fittings and how are they attached?

Answer

The most common types are JIC (37-degree flare), ORFS (O-ring face seal), BSPP (parallel thread), and NPT (tapered thread). Reusable fittings are screwed onto the hose and then inserted into a socket that is tightened to crimp the hose end. Permanently crimped fittings require a hydraulic hose crimping machine that presses a steel shell around the hose end to the manufacturer's die setting.

#315 · Fluid Power Systems

How do you safely replace a hydraulic hose in the field?

Answer

Isolate the circuit, relieve all pressure by cycling the controls with the pump off, then identify the correct replacement hose by matching the SAE rating, length, end fittings, and inside diameter. Clean the connection points before fitting, tighten fittings to specified torque (not just hand-tight), and check for leaks by cycling the system slowly before returning to full pressure. Never use your hand to find a hydraulic leak under pressure.

#316 · Fluid Power Systems

How do you replace the seals in a hydraulic cylinder?

Answer

Retract the cylinder fully to minimise oil spillage, isolate and de-energise the system, then remove the cylinder from the machine and drain the oil. Unscrew the gland nut or end cap to remove the piston rod assembly, then carefully remove all old seals and wiper rings using a non-metallic tool to avoid scoring the bore or rod. Install new seals from a matched seal kit, lubricating them with clean hydraulic oil, and reassemble to specified torque.

#317 · Fluid Power Systems

What is a pneumatic FRL unit and what does each section do?

Answer

FRL stands for Filter-Regulator-Lubricator, and it is a combined assembly that conditions compressed air before it reaches pneumatic equipment. The filter removes water, dirt, and oil aerosols from the incoming air. The regulator reduces line pressure to the working pressure required by the equipment, and the lubricator adds a fine mist of oil to lubricate internal pneumatic components.

#318 · Fluid Power Systems

How do you service a pneumatic FRL filter bowl?

Answer

Shut off the compressed air supply and bleed the downstream pressure before opening the filter bowl. Remove the bowl, empty any collected water and contaminants, and clean the bowl with a dry cloth or compressed air. Inspect the filter element for plugging or damage and replace it if the colour has changed or pressure drop is excessive.

#319 · Fluid Power Systems

How do you calculate the force output of a pneumatic cylinder?

Answer

Force equals supply pressure multiplied by the effective piston area. For example, a 50 mm bore cylinder at 600 kPa (6 bar) has an area of about 1,963 mm squared, giving a theoretical extend force of about 1,178 N. The actual force is slightly less due to seal friction, and the retract force is less because the rod reduces the effective area.

#320 · Fluid Power Systems

What are the main components of a compressed air supply system in a plant?

Answer

A compressed air system starts with a compressor (reciprocating, rotary screw, or centrifugal) that raises air pressure. Downstream comes an aftercooler that removes most of the water by cooling the compressed air, followed by an air dryer (refrigerant or desiccant type) to remove remaining moisture. A receiver tank stores compressed air to handle demand peaks, and a distribution system of pipes carries it to the plant.

#321 · Fluid Power Systems

What is the difference between a refrigerant air dryer and a desiccant air dryer?

Answer

A refrigerant dryer cools the compressed air to about 3 degrees C, causing moisture to condense and drain out; it produces air with a pressure dew point of about 3 degrees C, suitable for most plant air uses. A desiccant dryer uses a chemical bed (silica gel or activated alumina) to adsorb water vapour, achieving dew points as low as minus 40 degrees C for critical applications. Desiccant dryers require regeneration of the desiccant bed.

#322 · Fluid Power Systems

What is ISO 8573 compressed air quality classification?

Answer

ISO 8573 defines quality classes for compressed air based on solid particle content, water (dew point), and oil content, each rated by a number from 0 (cleanest) to 9. For example, a class 1.2.1 rating means very low particles, dew point below minus 40 degrees C, and total oil content below 0.01 mg/m³. The class required depends on the application: instrument air needs the highest quality.

#323 · Fluid Power Systems

What are the common types of pneumatic directional control valves?

Answer

Pneumatic valves are described by port count and position count just like hydraulic valves: a 5/2 valve has five ports and two positions and is the most common for controlling a double-acting cylinder (extend and retract). A 5/3 valve adds a centre position for mid-stroke holding. Pneumatic valves can be actuated by solenoids, manual operators, pilot air, or mechanical cams.

#324 · Fluid Power Systems

What is a quick-exhaust valve and why is it used on pneumatic cylinders?

Answer

A quick-exhaust valve is mounted directly at the cylinder port and vents exhaust air to atmosphere locally rather than through the long return line back to the valve. This greatly reduces back-pressure on the exhaust side, allowing much faster cylinder speed. It is used when high cycle speed is needed and the valve is mounted far from the cylinder.

#325 · Fluid Power Systems

What types of vacuum pumps are used in industrial systems?

Answer

Venturi (ejector) vacuum pumps use a jet of compressed air to create vacuum and have no moving parts, making them simple and maintenance-free but inefficient for large flow. Rotary vane vacuum pumps use sliding vanes in an off-centre rotor to pull air out mechanically, providing higher vacuum and flow rates. Liquid ring pumps use a rotating ring of water to compress and discharge gas, suited for wet or contaminated gas streams.

#326 · Fluid Power Systems

How do you size a vacuum cup for a pick-and-place application?

Answer

Calculate the force needed to hold the part by multiplying its weight by a safety factor of at least 2 (more for dynamic moves). Divide the required holding force in Newtons by the available vacuum level in kPa, then multiply by 10 to get the minimum cup area in square centimetres. Round up to the next standard cup size and verify that the cup material suits the part surface (smooth, porous, or oily).

#327 · Fluid Power Systems

What causes hydraulic pump cavitation and how is it prevented?

Answer

Hydraulic pump cavitation occurs when the inlet pressure drops below the vapour pressure of the oil, forming gas bubbles that collapse and erode the pump internals. Common causes are a clogged suction strainer, too small a suction line, oil that is too cold and viscous, or pump speed too high. Prevention includes keeping the suction strainer clean, using properly sized suction lines, and warming the oil before starting in cold weather.

#328 · Fluid Power Systems

What is aeration in a hydraulic system and how is it identified?

Answer

Aeration is free air or gas mixed into the hydraulic fluid, caused by air entering through leaky suction fittings, low oil level, or excessive turbulence in the tank. The symptoms are a milky or foamy oil appearance, spongy actuator movement, and a knocking or rattling noise from the pump. Fix by finding and sealing the air ingestion point, correcting oil level, and allowing the system to degas at idle.

#329 · Fluid Power Systems

How does hydraulic fluid viscosity affect system performance?

Answer

If viscosity is too low (too thin), internal pump leakage increases, lubrication film breaks down, and the system overheats. If viscosity is too high (too thick), pump cavitation is more likely at start-up, pressure drops across filters and valves increase, and response becomes sluggish. Most hydraulic systems use ISO VG 46 oil, which is designed to stay in the correct viscosity range over the normal operating temperature band.

#330 · Fluid Power Systems

What is the purpose of a hydraulic reservoir and what features does it need?

Answer

The reservoir stores the system's oil supply and allows heat to dissipate through the tank walls, gives entrained air time to rise and escape, and lets contaminants settle to the bottom. A properly designed reservoir has a baffle plate between the return and suction ports to prevent return oil from going directly back to the pump. It also needs a breather filter, a drain plug at the lowest point, and an oil level gauge.

#331 · Fluid Power Systems

What is a hydraulic power unit (HPU) and what are its main components?

Answer

An HPU is a self-contained assembly that provides pressurised hydraulic fluid to a machine or system. Its main components are a reservoir, electric motor, pump, relief valve, pressure gauge, suction strainer, return filter, and often a cooler and accumulator. The HPU is a standardised module that can be installed near or remote from the actuators it serves.

#332 · Fluid Power Systems

How do you commission a new hydraulic system for the first time?

Answer

Fill the reservoir with clean, filtered oil and prime the pump by filling its case before starting. Start the system with the relief valve at a low setting and cycle all actuators slowly at low pressure to flush debris from the lines. Gradually increase the relief valve setting to the working pressure, check all connections for leaks, then change the return filter after the initial flush period.

#333 · Fluid Power Systems

What is the purpose of case drain lines on hydraulic piston pumps and motors?

Answer

Piston pumps and motors have internal leakage that fills the case housing with low-pressure oil; the case drain line returns this oil to the tank. If the case drain is blocked or back-pressured, case pressure rises and can blow out shaft seals or damage the barrel and pistons. The case drain must never be restricted and should have a check valve if there is any risk of back-pressure from the tank return line.

#334 · Fluid Power Systems

What is meant by the differential cylinder speed ratio?

Answer

The extend and retract speeds of a double-acting cylinder are different because the effective piston area is different on each side. The retract speed is faster than the extend speed when the same flow is applied, because the oil must fill a smaller volume (cap area minus rod area) on the rod side. The speed ratio equals the cap bore area divided by the annular area on the rod side.

#335 · Fluid Power Systems

What is a hydraulic intensifier and when is it used?

Answer

A hydraulic intensifier (booster) takes low-pressure supply oil and produces a smaller volume of higher-pressure oil by using the ratio of two different piston areas. For example, a 4:1 intensifier converts 100 bar input to 400 bar output at one-quarter the flow rate. Intensifiers are used for clamping, pressing, and spot welding applications where very high pressure is needed briefly without a high-pressure pump.

#336 · Fluid Power Systems

What is the purpose of a drain valve (dump valve) in a hydraulic circuit?

Answer

A dump valve allows the stored energy in an accumulator or pressurised circuit to be safely discharged to tank before maintenance begins. It is normally a manual ball valve or a solenoid valve wired to de-energise when the machine stops. Lockout/tagout procedures must include confirming that accumulators are discharged before opening any connections.

#337 · Fluid Power Systems

What is the difference between meter-in and meter-out flow control?

Answer

Meter-in places the flow control on the supply line entering the actuator, controlling the rate at which oil enters and limiting extend speed. Meter-out places the flow control on the exhaust line leaving the actuator, restricting flow out and controlling speed by creating back-pressure. Meter-out is preferred for overrunning loads (such as a cylinder being pushed by gravity) because it keeps the actuator under back-pressure and prevents uncontrolled runaway.

#338 · Fluid Power Systems

What safety precautions must be followed before working on any hydraulic circuit?

Answer

Apply lockout/tagout to the electric motor driving the pump, then cycle the directional control valve in both directions to vent trapped pressure. Use a calibrated pressure gauge to confirm zero pressure at all test points, including the accumulator, before disconnecting any fitting. Wear eye and face protection, because even residual pressure in a small fitting can cause oil injection injuries.

#339 · Fluid Power Systems

What is hydraulic shock (water hammer) and how is it reduced?

Answer

Hydraulic shock is a pressure spike caused by suddenly stopping a moving column of hydraulic oil, such as when a solenoid valve closes instantly. The pressure spike can exceed the relief valve setting momentarily and damage valves, fittings, or cylinders. It is reduced by using deceleration valves, soft-start solenoids with controlled ramp timing, or accumulators mounted close to the actuator.

#340 · Fluid Power Systems

How do you troubleshoot a hydraulic cylinder that drifts under load?

Answer

Drifting means oil is leaking past a seal, either internally across the piston or externally past a valve. First check if the directional control valve is holding pressure by isolating it; if the cylinder still drifts, the piston seal is bypassing. If the cylinder holds when the valve is isolated, the valve spool is worn or contaminated. Worn counterbalance valves or pilot-operated check valves can also allow drift.

#341 · Fluid Power Systems

What does high system temperature indicate about a hydraulic circuit?

Answer

Excessive heat indicates that more power is being converted to heat than the cooler can remove, which is caused by pressure drops across partially open valves, continuous relief valve bypass, or high internal leakage in worn components. Check the relief valve setting is not too close to the working pressure, check for stuck or partially shifted valves, and confirm the cooler is functioning. An oversized relief valve bypass wastes energy as heat continuously.

#342 · Fluid Power Systems

What is the function of a hydraulic tank breather filter?

Answer

As oil level in the tank rises and falls with actuator movement, air must flow in and out of the tank through the breather. The breather filter prevents atmospheric dust and moisture from entering the reservoir with this air, protecting the oil and system cleanliness. Breather filters should be rated to at least the same cleanliness level as the system's return filter.

#343 · Fluid Power Systems

What is pump efficiency and why does it matter for system design?

Answer

Pump efficiency is the ratio of hydraulic power output to mechanical power input. Volumetric efficiency measures how much of the theoretical displacement actually leaves the pump (losses from internal leakage). Overall efficiency multiplies volumetric and mechanical efficiencies together to give the total power conversion ratio. A pump with 85 percent overall efficiency requires 18 percent more input power than the purely theoretical calculation.

#344 · Fluid Power Systems

What causes a hydraulic pump to lose displacement over time?

Answer

Internal wear between the pump's rotating components and their mating surfaces (gear tip-to-bore clearance, vane tip, or piston barrel-to-valve plate) increases as the pump ages, allowing more internal leakage (called slip). This reduces volumetric efficiency, meaning less flow is delivered at the same speed. Contaminated oil accelerates this wear, which is why cleanliness and proper filtration directly extend pump life.

#345 · Fluid Power Systems

How does a pneumatic rotary actuator work and where is it used?

Answer

A pneumatic rotary actuator converts air pressure into rotational motion using a rack-and-pinion or vane design. In a rack-and-pinion type, a cylinder pushes a rack that rotates a pinion gear on the output shaft; typical rotation is 90 or 180 degrees. They are used to operate quarter-turn ball valves, butterfly valves, and grippers on automated assembly lines.

#346 · Fluid Power Systems

What is a two-stage hydraulic pump circuit (hi-lo circuit) and why is it used?

Answer

A hi-lo circuit uses two pumps in parallel: a large high-flow pump and a small high-pressure pump. On the advance stroke, both pumps combine their flow for fast movement at low pressure. When the load is met and pressure rises, an unloading valve dumps the large pump back to tank, leaving only the small pump to do the work at high pressure. This reduces heat and saves energy compared to running a large pump at high pressure constantly.

#347 · Fluid Power Systems

What is a hydraulic manifold and what are its advantages?

Answer

A hydraulic manifold is a machined block with internal passages that connect valves and components, replacing individual pipes and fittings between them. Manifolds reduce the number of potential leak points, save installation space, and make the circuit easier to service because valves are grouped together. Custom manifolds are designed using CAD to route passages without cross-connections.

#348 · Fluid Power Systems

What is the purpose of a flow divider in a hydraulic circuit?

Answer

A flow divider splits incoming flow into two or more equal streams so that multiple actuators move in synchronisation regardless of load differences between them. Gear-type flow dividers use synchronized gears to meter equal volumes. They are used in applications like levelling platforms and side-by-side cylinders that must extend at the same rate.

#349 · Fluid Power Systems

How do you read a hydraulic circuit diagram to trace a cylinder extend sequence?

Answer

Start at the hydraulic power unit symbol, then follow the pressure line (shown as a solid line) to the directional control valve. Identify which valve position is selected for extend (usually shown by an arrow inside a box crossing from P to A and B to T). Trace the flow from port A to the cap end of the cylinder symbol, and confirm the rod end connects to port B returning to T (tank).

#350 · Fluid Power Systems

What is the function of a shuttle valve in a pneumatic or hydraulic circuit?

Answer

A shuttle valve has two inlets and one outlet; it automatically connects whichever inlet has the higher pressure to the outlet while blocking the lower-pressure inlet. It is used to allow either of two control signals (for example, local and remote control) to actuate a valve without the two control lines feeding back into each other. Shuttle valves are also used in brake circuits to pass the higher of two competing pilot signals.

#351 · Preventive and Predictive Maintenance

What is the difference between preventive maintenance and predictive maintenance?

Answer

Preventive maintenance (PM) is performed on a fixed time or usage schedule, regardless of the equipment's actual condition, such as changing oil every 2,000 hours. Predictive maintenance (PdM) uses condition-monitoring data (vibration, oil analysis, temperature) to predict when a component is likely to fail and schedules work just before that point. PdM reduces unnecessary part replacements and catches failures that fixed schedules would miss.

#352 · Preventive and Predictive Maintenance

What is the difference between time-based and condition-based maintenance strategies?

Answer

Time-based maintenance replaces or services components at fixed intervals regardless of condition, which is simple to schedule but can result in replacing parts that still have useful life. Condition-based maintenance monitors the actual state of a component and only triggers action when a defined threshold is reached or exceeded. Condition-based maintenance is more efficient but requires monitoring equipment and expertise to interpret the data.

#353 · Preventive and Predictive Maintenance

What is Reliability-Centred Maintenance (RCM) and what is its goal?

Answer

RCM is a systematic process for determining the most cost-effective maintenance strategy for each piece of equipment based on its function, failure modes, and the consequences of failure. It asks seven structured questions about each asset to decide whether PM, PdM, or run-to-failure is the right approach. The goal is to preserve equipment function at the lowest total cost while managing safety and production risk.

#354 · Preventive and Predictive Maintenance

What is Failure Mode and Effects Analysis (FMEA) and how is it used in maintenance?

Answer

FMEA is a structured method for identifying all the ways a component or system can fail (failure modes), what causes each failure, and what the effect on the process or safety would be. Each failure mode is rated for severity, frequency, and detectability to calculate a Risk Priority Number (RPN). Maintenance tasks are then designed to address the highest-RPN failure modes first.

#355 · Preventive and Predictive Maintenance

What are MTBF and MTTR and how are they used in maintenance planning?

Answer

MTBF (Mean Time Between Failures) is the average operating time between failures of a repairable item; a higher MTBF means more reliable equipment. MTTR (Mean Time To Repair) is the average time it takes to restore equipment to service after a failure; a lower MTTR means faster recovery. Maintenance programs aim to maximise MTBF through better lubrication and alignment, and minimise MTTR through parts availability and trained technicians.

#356 · Preventive and Predictive Maintenance

What do the vibration frequency harmonics 1x, 2x, and 0.5x indicate in a spectrum?

Answer

1x (one times running speed) is the fundamental frequency and is usually related to imbalance or misalignment. 2x running speed indicates angular misalignment, looseness, or a bent shaft. 0.5x (one-half running speed) is a sub-harmonic that typically indicates oil film instability (oil whirl) in sleeve bearings or rub conditions.

#357 · Preventive and Predictive Maintenance

What does ISO 10816 specify and how is it used for vibration severity?

Answer

ISO 10816 provides vibration severity limits for measuring vibration in the non-rotating parts (bearing housings) of rotating machines. It divides machines into groups by size and mounting type and defines four zones from Zone A (new machine, acceptable) to Zone D (dangerous, immediate shutdown required). A millwright uses these limits to classify whether a machine's vibration level is acceptable, marginal, or requiring action.

#358 · Preventive and Predictive Maintenance

What oil analysis parameters are most important for a hydraulic system?

Answer

Viscosity tells you if the oil has thinned (thermal degradation or contamination) or thickened (oxidation). The acid number (AN or TAN) rises as the oil oxidises and forms acids that corrode metal surfaces. Particle count (ISO 4406 code) shows contamination level. Water content (in ppm) indicates if moisture has entered the system, which can cause rust, microbial growth, and additive depletion.

#359 · Preventive and Predictive Maintenance

How is infrared thermography used as a predictive maintenance tool?

Answer

A thermal camera detects heat patterns on equipment surfaces, revealing hot spots caused by overloaded bearings, loose electrical connections, misalignment, or blocked cooling passages. Thermography is non-contact and can be performed on running equipment, making it fast and safe. Regular thermal surveys establish a baseline so that developing problems are caught before they cause a failure.

#360 · Preventive and Predictive Maintenance

How is ultrasonic testing used to find bearing defects and compressed air leaks?

Answer

An ultrasonic detector picks up high-frequency sound (typically 38-40 kHz) that is beyond human hearing. Bearing defects produce a specific ultrasonic signature before they become audible, allowing early detection. Compressed air and gas leaks generate turbulent ultrasonic noise at the leak point, so the detector can find small leaks in noisy plant environments quickly.

#361 · Preventive and Predictive Maintenance

What is the soft foot tolerance limit for precision machinery shaft alignment?

Answer

The generally accepted limit for residual soft foot before proceeding with shaft alignment is 0.05 mm (0.002 inches) total indicator reading per foot. Any foot showing more movement than this when its bolt is loosened must be shimmed before alignment is started. Correcting soft foot before aligning is not optional; even perfect alignment will be lost immediately when bolts are tightened on a machine with uncorrected soft foot.

#362 · Preventive and Predictive Maintenance

How do you determine the correct re-grease interval and volume for a bearing?

Answer

Consult the bearing manufacturer's lubrication guide, which gives intervals based on bearing bore diameter, RPM, and operating temperature. A common formula for grease volume is 0.005 times the bearing outside diameter (mm) times its width (mm), giving the result in grams. Re-grease while the machine is running when possible, so fresh grease purges old grease through the relief fitting.

#363 · Preventive and Predictive Maintenance

What are the criteria for replacing a rolling element bearing during maintenance?

Answer

Replace a bearing if vibration analysis shows sub-surface fatigue signatures in the high-frequency range, if ultrasonic testing shows crackling or impulse noise, or if the bearing temperature runs consistently higher than the normal baseline. Physical criteria for replacement include pitting, spalling, or scoring on the raceways or rolling elements, any cracking of the inner or outer ring, and cage damage found during inspection.

#364 · Preventive and Predictive Maintenance

What information is recorded on a maintenance work order?

Answer

A work order records the equipment ID and location, the type and description of the work to be performed, any parts or materials required, the estimated and actual labour hours, the completion date, and the name of the technician who did the work. It also captures any findings, measurements taken (alignment readings, clearances), and any follow-up work identified. Work orders are the foundation of the maintenance history record for each asset.

#365 · Preventive and Predictive Maintenance

What is included in a machine shutdown checklist before major maintenance?

Answer

A shutdown checklist covers: notifying operations and receiving a work permit, isolating all energy sources (electrical, pneumatic, hydraulic, steam) and applying lockout/tagout, draining fluids as required, allowing hot parts to cool, and confirming zero energy state with a test. It also identifies any special hazards such as confined space entry, elevated work, or hazardous materials before work begins.

#366 · Preventive and Predictive Maintenance

What is a startup checklist and why is it used after a major machine overhaul?

Answer

A startup checklist confirms that all maintenance tasks are complete, all guards and covers are reinstalled, all tools and parts are cleared from the machine, fluids are at correct levels, and all lockout devices have been removed by the authorised workers. It then guides a controlled first start: run at no load first, check for leaks or unusual noise, confirm temperatures and pressures are normal before returning to full production load. Skipping the startup checklist risks damage from assembly errors or missing steps.

#367 · Preventive and Predictive Maintenance

What is a lubrication program and what does it include?

Answer

A lubrication program is a documented system that lists every lubrication point on every machine, the correct lubricant type and grade, the quantity to apply, and the frequency. It assigns routes to technicians and tracks completion. A good program also includes oil sampling schedules for condition monitoring and prevents mixing incompatible grease types by clearly labelling fitting colours.

#368 · Preventive and Predictive Maintenance

What are the consequences of over-greasing a rolling element bearing?

Answer

Packing too much grease into a bearing housing causes the rolling elements to churn through the excess grease, generating heat. The heat raises operating temperature, accelerates oxidation of the grease, and can cause the grease to liquefy and leak out of the seal. Over-greasing is one of the most common causes of premature bearing failure, even though it feels like doing extra maintenance.

#369 · Preventive and Predictive Maintenance

What is vibration amplitude and what units are used at different frequency ranges?

Answer

Vibration amplitude measures the severity of the motion; the unit used depends on the frequency range of interest. Displacement (in micrometres or thousandths of an inch) is used for low frequencies below 10 Hz. Velocity (in mm/s) is the standard unit for the mid-range (10 to 1,000 Hz) used in most ISO severity standards. Acceleration (in g) is used for high frequencies above 1,000 Hz, such as bearing defect frequencies.

#370 · Preventive and Predictive Maintenance

What does a rising trend in oil acid number (TAN) indicate?

Answer

A rising Total Acid Number (TAN) means the oil is oxidising and forming organic acids as its antioxidant additives are depleted. Acidic oil attacks and corrodes metal surfaces inside pumps, valves, and cylinders. When the TAN rises to double the new oil baseline or exceeds the condemning limit stated by the oil supplier, the oil must be changed before corrosive damage occurs.

#371 · Preventive and Predictive Maintenance

What is the P-F interval in condition-based maintenance?

Answer

The P-F interval is the time between the point where a potential failure (P) can first be detected by a monitoring method and the point where it becomes a functional failure (F). A long P-F interval gives maintenance teams plenty of time to plan and schedule corrective work. The monitoring frequency must be shorter than the P-F interval to reliably catch failures before they become critical.

#372 · Preventive and Predictive Maintenance

How does motor current analysis help detect mechanical problems in a machine?

Answer

Motor current signature analysis (MCSA) monitors the current drawn by an electric motor driving a machine. Mechanical faults such as eccentricity, broken rotor bars, or load variations from a failing gearbox or pump create small periodic variations in the motor current that can be detected as sidebands around the supply frequency. MCSA can detect problems in the driven equipment without any sensors on the machine itself.

#373 · Preventive and Predictive Maintenance

What is a baseline vibration reading and why is it taken on new equipment?

Answer

A baseline (or reference) vibration reading is taken on a new or freshly overhauled machine under normal operating conditions and documents the vibration signature when the machine is in known good condition. All future readings are compared to this baseline to identify changes, because a slight increase in a specific frequency component is far more meaningful than the absolute level alone. Without a baseline, it is difficult to judge whether current readings are normal for that specific machine.

#374 · Preventive and Predictive Maintenance

What is the purpose of an oil flushing procedure when commissioning new equipment?

Answer

An oil flush circulates clean oil through the system at high velocity and temperature before the machine is put into service, to remove metal particles from machining, pipe scale, welding slag, and assembly lubricants that contaminate the fresh oil fill. The flush oil is sampled until it meets the target cleanliness level before the machine starts normal operation. Skipping the flush is a common cause of early bearing and pump failure on new installations.

#375 · Preventive and Predictive Maintenance

How do you interpret a bearing defect frequency in a vibration spectrum?

Answer

Each rolling element bearing has four characteristic defect frequencies (BPFO, BPFI, BSF, FTF) calculated from its geometry and running speed. If a peak appears in the vibration spectrum at one of these calculated frequencies, it indicates that particular bearing race or rolling element has a developing defect. Sidebands around the defect frequency indicate the defect is more advanced and is being modulated by shaft rotation.

#376 · Preventive and Predictive Maintenance

What is a borescope used for in equipment inspection?

Answer

A borescope is a thin, flexible or rigid tube with a camera that lets you see inside engines, gearboxes, and pipes without taking them apart. It helps find cracks, wear, carbon buildup, or foreign objects. This saves time and avoids unnecessary disassembly.

#377 · Preventive and Predictive Maintenance

How does a stroboscope help inspect rotating equipment while running?

Answer

A stroboscope flashes a light at the same speed as a rotating part, making it appear frozen in place. This lets you inspect pulleys, couplings, and fans for wobble, cracks, or imbalance while they are still spinning. You adjust the flash rate until the part looks still.

#378 · Preventive and Predictive Maintenance

Why is laser alignment considered a predictive maintenance tool?

Answer

Laser alignment measures misalignment precisely, and trending those measurements over time shows if a machine is shifting or settling on its foundation. Catching misalignment early prevents bearing and coupling failures before they cause downtime. This makes it both a corrective and a predictive tool.

#379 · Preventive and Predictive Maintenance

What do angular wear patterns on a jaw coupling indicate?

Answer

Uneven wear on one side of the jaw coupling spider indicates angular misalignment between the driver and driven shafts. If all jaws are worn evenly but heavily, the coupling is overloaded or the wrong durometer spider was selected. Identifying the wear pattern guides the correct repair.

#380 · Preventive and Predictive Maintenance

How does an infrared thermometer monitor bearing temperature in the field?

Answer

An infrared thermometer reads surface temperature without contact by measuring radiated heat from the bearing housing. A sudden rise above the baseline temperature, typically more than 10 to 15 degrees Celsius, signals a lubrication problem or early bearing damage. You record readings regularly to build a trend.

#381 · Preventive and Predictive Maintenance

What is an RTD and how is it used for bearing temperature monitoring?

Answer

An RTD (resistance temperature detector) is a sensor embedded in or bolted to a bearing housing that changes electrical resistance with temperature. It feeds continuous data to a control system or data logger, allowing real-time monitoring and automatic alarms. RTDs are more accurate than thermocouples for this application.

#382 · Preventive and Predictive Maintenance

What does bearing enveloping analysis detect that overall vibration misses?

Answer

Enveloping (also called high-frequency demodulation or spike energy) filters out low-frequency machine noise and looks at the high-frequency impact pulses created when a bearing defect strikes a race. These pulses appear long before overall vibration rises, giving early warning of bearing damage. It is one of the most sensitive bearing fault detection methods.

#383 · Preventive and Predictive Maintenance

Name four items on a gearbox inspection checklist during a shutdown.

Answer

A standard gearbox inspection checks oil level and condition (colour, smell, particle content), gear tooth contact pattern and wear, bearing end-play and radial clearance, and seal condition for leaks. You also check the breather to confirm it is clean and functional. Any findings are recorded in the equipment history.

#384 · Preventive and Predictive Maintenance

What is a fishbone diagram and when is it used in root cause analysis?

Answer

A fishbone diagram (also called Ishikawa or cause-and-effect diagram) organizes possible causes of a failure into categories such as machine, method, material, man, measurement, and environment. The effect (the failure) goes on the right, and branches show potential causes. It is used during a root cause analysis meeting to make sure no cause is overlooked.

#385 · Preventive and Predictive Maintenance

Explain the 5 Why root cause analysis method with a simple example.

Answer

The 5 Why method asks why a failure happened, then asks why that answer happened, repeating up to five times until you reach the root cause. For example: bearing failed, why? No grease. Why? Grease gun empty. Why? No reorder process. The root cause is a missing procedure, not just a dry bearing. Fixing the process prevents recurrence.

#386 · Preventive and Predictive Maintenance

How often should a torque wrench be calibrated and why does it matter?

Answer

Most manufacturers recommend calibrating torque wrenches every 12 months or after 5,000 cycles, whichever comes first. An out-of-calibration wrench can under-torque fasteners, causing loosening, or over-torque them, causing fatigue failures. Calibration records must be kept to prove maintenance work was done to the correct specification.

#387 · Preventive and Predictive Maintenance

How does a desiccant breather prevent lubrication contamination in a gearbox?

Answer

As a gearbox heats and cools, it breathes air in and out through its breather. A desiccant breather contains silica gel and a filter that remove moisture and particulates from the incoming air before they reach the oil. This is one of the most cost-effective ways to extend oil and bearing life.

#388 · Preventive and Predictive Maintenance

What is an oil sampling port and how is it used correctly?

Answer

An oil sampling port is a dedicated valve fitted to a gearbox or reservoir that lets you draw a mid-stream oil sample without shutting down or opening the case. Samples are taken with a clean syringe or pump at the same location and operating temperature every time for consistent trending. The sample is sent to a lab for wear metal, viscosity, and contamination analysis.

#389 · Preventive and Predictive Maintenance

What are the three types of triggers that can start a planned maintenance interval?

Answer

Planned maintenance intervals can be triggered by time (calendar-based, such as every 90 days), by usage (hours-based, such as every 2,000 operating hours), or by condition (condition-based, such as when vibration exceeds a set threshold). Most industrial PM programs use a combination of all three triggers depending on the equipment criticality.

#390 · Preventive and Predictive Maintenance

What does OEE stand for and what does it measure?

Answer

OEE stands for Overall Equipment Effectiveness. It combines three factors: availability (was the machine running when needed?), performance (was it running at full speed?), and quality (was it producing good parts?). Multiplying all three gives a single percentage that shows how efficiently equipment is being used.

#391 · Preventive and Predictive Maintenance

How is equipment availability calculated as a maintenance metric?

Answer

Availability equals uptime divided by total planned operating time, expressed as a percentage. For example, if a machine was planned to run 40 hours but was down for 4 hours, availability is 90 percent. Tracking availability over time shows whether maintenance work is improving or allowing decline in reliability.

#392 · Preventive and Predictive Maintenance

What does a portable vibration analyzer measure and what data does it collect?

Answer

A portable vibration analyzer uses an accelerometer held or mounted on a bearing housing to measure vibration amplitude and frequency. It collects overall vibration levels, spectrum data (FFT), and time waveform signals. This data is compared to baseline readings and alarm limits to identify faults such as imbalance, misalignment, looseness, or bearing damage.

#393 · Preventive and Predictive Maintenance

How does an ultrasound gun detect early bearing faults and air leaks?

Answer

An ultrasound gun detects high-frequency sound (typically 20 to 100 kHz) that is inaudible to the human ear. A failing bearing produces a distinctive crackling or rushing sound at these frequencies well before any temperature rise or low-frequency vibration change. The same tool finds compressed air and steam leaks by detecting the ultrasound they emit.

#394 · Preventive and Predictive Maintenance

What does an infrared camera show during electrical and mechanical inspections?

Answer

An infrared camera creates a thermal image showing temperature differences across a surface. In electrical inspections, hot spots indicate loose connections or overloaded components. In mechanical inspections, a hot bearing housing or overheating coupling shows up as a bright area, allowing the problem to be found without shutting down the equipment.

#395 · Preventive and Predictive Maintenance

What is a predictive maintenance database used for in a mill or plant?

Answer

A predictive maintenance database stores equipment readings (vibration, temperature, oil analysis results) over time so trends can be identified. When a reading drifts from its historical normal range, it triggers an alert before failure occurs. The database also stores baseline readings taken on new or freshly rebuilt equipment.

#396 · Preventive and Predictive Maintenance

Why is trending data more useful than a single vibration reading?

Answer

A single reading only tells you the current state. Trending compares readings over weeks or months to show whether a machine is getting better, staying stable, or slowly getting worse. A gradual rise in vibration over several months points to developing imbalance or bearing wear, allowing maintenance to be scheduled before breakdown.

#397 · Preventive and Predictive Maintenance

What information should be recorded in an equipment history file?

Answer

Equipment history records should include every maintenance action performed, the date and who did it, parts replaced (brand, model, lot number), lubricants added (type and quantity), vibration and temperature readings, and any failures or findings. This history is what makes root cause analysis possible and helps justify capital replacement decisions.

#398 · Preventive and Predictive Maintenance

What coupling wear pattern indicates parallel (offset) misalignment?

Answer

Parallel misalignment causes wear across the full width of both coupling hubs but concentrated on opposite sides when the two hubs are compared. In a gear coupling, the gear teeth show wear on one flank only. In a jaw coupling, all spider legs wear but one side of the bore shows more loading.

#399 · Preventive and Predictive Maintenance

What vibration frequency signature indicates mechanical looseness on an FFT spectrum?

Answer

Mechanical looseness (such as a loose bearing housing or foot) appears as a series of harmonics at 1X, 2X, 3X, and higher multiples of running speed on the FFT spectrum. The pattern often has many peaks and can look like a forest of lines. Phase analysis can confirm looseness by checking for unstable phase readings.

#400 · Preventive and Predictive Maintenance

What is the difference between failure mode and root cause in RCA?

Answer

The failure mode is the way the component failed, for example a bearing inner race crack. The root cause is the underlying reason it happened, such as inadequate lubrication due to a clogged grease line. Fixing only the failure mode (replacing the bearing) without fixing the root cause means the same failure will repeat.

#401 · Preventive and Predictive Maintenance

What does MTBF mean and how is it calculated?

Answer

MTBF stands for Mean Time Between Failures. It is calculated by dividing total operating time by the number of failures in that period. For example, if a pump ran 4,000 hours and failed 4 times, MTBF is 1,000 hours. A rising MTBF means reliability is improving; a falling MTBF signals a problem.

#402 · Preventive and Predictive Maintenance

What is the purpose of an oil cleanliness code (ISO 4406) in a lubrication program?

Answer

ISO 4406 counts the number of particles per millilitre in oil at three sizes (4, 6, and 14 microns) and assigns a three-number code such as 18/16/13. Cleaner oil has lower code numbers. Hydraulic systems and precision gearboxes require specific cleanliness levels, and lab analysis confirms whether those targets are being met.

#403 · Preventive and Predictive Maintenance

How does spectrographic oil analysis detect internal machine wear?

Answer

Spectrographic analysis measures the concentration of metal particles dissolved or suspended in the oil sample, reported in parts per million. Rising iron indicates steel wear, rising copper points to bronze bushings or thrust washers, and rising lead suggests babbit or bearing overlay wear. Trending these numbers over multiple samples shows whether wear is normal or accelerating.

#404 · Preventive and Predictive Maintenance

What is a P-F curve and how does it guide maintenance scheduling?

Answer

The P-F curve shows the timeline between the point where a potential failure (P) becomes detectable and the point of functional failure (F). The gap between P and F is the P-F interval. Condition monitoring tools such as vibration analysis or oil sampling must be used at intervals shorter than the P-F interval to catch the fault in time to plan repairs.

#405 · Preventive and Predictive Maintenance

What is a criticality ranking and why does it affect PM frequency?

Answer

Criticality ranking scores equipment based on the consequence of its failure: safety risk, production loss, repair cost, and availability of a spare. High-criticality equipment (such as the only pump feeding a process line) gets more frequent inspections and more thorough PM tasks. Low-criticality equipment with an easy-to-replace spare may only need calendar-based checks.

#406 · Common Occupational Skills

How do you convert a fraction like $\frac{3}{8}$ to a decimal for a measurement?

Answer

Divide the top number (numerator) by the bottom number (denominator): 3 divided by 8 equals 0.375. This is useful when a drawing calls out a fractional dimension but your micrometer reads in decimals. Practice the common fractions: $\frac{1}{4}$ is 0.250, $\frac{1}{2}$ is 0.500, $\frac{3}{4}$ is 0.750.

#407 · Common Occupational Skills

What is the ratio and proportion method used for in millwright calculations?

Answer

Ratio and proportion lets you solve for an unknown when you know three of four related values. For example, if a 20-tooth gear drives a 60-tooth gear and the driver turns at 900 RPM, you set up $20/60$ equals $x/900$ and solve: the driven gear turns at 300 RPM. This method is used constantly for gear ratios, pulley sizes, and flow rates.

#408 · Common Occupational Skills

How do you calculate what percentage one number is of another?

Answer

Divide the part by the whole, then multiply by 100. For example, if a shaft is 1.998 inches and the nominal size is 2.000 inches, the shaft is 1.998 divided by 2.000 times 100, which equals 99.9 percent of nominal. Percentage calculations are used for efficiency ratings, torque capacity, and load checks.

#409 · Common Occupational Skills

What is the formula for the area of a circle, and where is it used in the trade?

Answer

Area equals pi times radius squared ($A = \pi \times r^2$). You use this when calculating the cross-sectional area of a shaft, pipe bore, or piston to find force, stress, or flow area. For a 4-inch diameter pipe, the radius is 2 inches, so the area is 3.1416 times 2 squared (which equals 3.1416 times 4), giving about 12.57 square inches.

#410 · Common Occupational Skills

How do you calculate the volume of a cylinder such as a hydraulic ram?

Answer

Volume equals pi times radius squared times length ($V = \pi \times r^2 \times L$). For a hydraulic cylinder with a 3-inch bore and 12-inch stroke, volume is 3.1416 times 1.5 squared times 12, which equals about 84.8 cubic inches. Convert to gallons by dividing by 231 cubic inches per gallon to find oil displacement.

#411 · Common Occupational Skills

How do you convert inches to millimetres for a metric drawing dimension?

Answer

Multiply inches by 25.4 to get millimetres. For example, 2.500 inches times 25.4 equals 63.5 mm. To go the other way, divide millimetres by 25.4. Keeping a conversion card handy or memorizing that one inch equals 25.4 mm is essential when working from both imperial and metric drawings.

#412 · Common Occupational Skills

How do you convert PSI to kilopascals for a metric pressure specification?

Answer

Multiply PSI by 6.895 to get kilopascals (kPa). For example, 100 PSI equals 689.5 kPa. To convert back, divide kPa by 6.895. One atmosphere is 14.696 PSI or approximately 101.3 kPa. This conversion comes up when setting relief valves or reading hydraulic specifications written in different unit systems.

#413 · Common Occupational Skills

How do you convert foot-pounds of torque to Newton-metres?

Answer

Multiply foot-pounds by 1.356 to get Newton-metres (Nm). For example, 100 ft-lb times 1.356 equals 135.6 Nm. To go the other way, divide Nm by 1.356. Torque specifications on modern equipment often appear in both units, and mixing them up is a serious error when torquing fasteners.

#414 · Common Occupational Skills

What is the sine ratio in a right triangle and how is it used for angle work?

Answer

Sine equals the length of the opposite side divided by the hypotenuse (SOH). A millwright uses this to calculate the rise of an inclined machine base, the height a sling leg reaches above a load, or the offset in a bend. If the angle is 30 degrees, the sine is 0.500, meaning the opposite side is half the hypotenuse.

#415 · Common Occupational Skills

What is the Pythagorean theorem and when does a millwright use it?

Answer

The Pythagorean theorem states that in a right triangle, the hypotenuse squared equals the sum of the other two sides squared ($c^2 = a^2 + b^2$). A millwright uses it to check that a machine base is square (the 3-4-5 method), to find the diagonal of a rectangular layout, or to calculate the length of a diagonal brace.

#416 · Common Occupational Skills

What does orthographic projection show on a mechanical drawing?

Answer

Orthographic projection shows an object in two or more flat views (front, top, and side) at exactly 90 degrees to each other. Features visible in one view align with the same feature in the adjacent view. A millwright reads these views together to understand the full 3D shape of a part before fabricating or machining it.

#417 · Common Occupational Skills

What is an isometric view on a drawing and when is it used?

Answer

An isometric view shows a 3D-looking picture of a part or assembly drawn on a flat page with all three axes at 30 degrees to the horizontal. It is used for installation drawings, piping layouts, and assembly guides because it is easier to visualize than multi-view drawings. Dimensions cannot be scaled directly from an isometric view.

#418 · Common Occupational Skills

What does a section view on a drawing reveal about a part?

Answer

A section view shows what the inside of a part looks like as if it were cut in half and one piece removed. Cross-hatching shows the solid material that was cut through. Section views are used to show internal bores, keyways, wall thickness, and features that would be hidden lines in a standard view.

#419 · Common Occupational Skills

What information is found in a drawing title block?

Answer

The title block contains the part name and number, the drawing number, the scale, the material specification, the tolerance block (default tolerances if not stated on the drawing), the company name, and the revision level. Always check the title block before reading a drawing to confirm you have the correct and latest version.

#420 · Common Occupational Skills

What does a revision block on a drawing tell you?

Answer

The revision block lists every change made to the drawing after it was first issued, with a revision letter (A, B, C...), the date of the change, a brief description of what changed, and who approved it. Always use the highest revision level available to avoid working from an outdated drawing.

#421 · Common Occupational Skills

What does a fillet weld symbol on a drawing indicate to a millwright?

Answer

A fillet weld symbol is a triangle on the reference line of a welding symbol. It tells you the weld is a triangular cross-section bead joining two surfaces at an angle, typically a T-joint or lap joint. The number next to the triangle is the leg size in inches or millimetres, and its position above or below the line indicates which side of the joint gets welded.

#422 · Common Occupational Skills

What does the flatness GD&T; symbol control on a machined part?

Answer

Flatness controls how much a surface can deviate from a perfectly flat plane. The tolerance zone is two parallel flat planes that the entire surface must lie between. It is used for precision machine bases, mounting pads, and split bearing housings where a flat mating surface is critical for proper assembly and alignment.

#423 · Common Occupational Skills

What does the circularity (roundness) GD&T; symbol control?

Answer

Circularity controls how close each cross-section of a round feature is to a perfect circle. The tolerance zone is two concentric circles that every point on the cross-section must fall between. It is important for shaft journals, bearing seats, and cylinder bores where out-of-round surfaces cause uneven loading and premature wear.

#424 · Common Occupational Skills

What is the difference between total runout and circular runout in GD&T;?

Answer

Circular runout measures the variation at each individual cross-section as a shaft is rotated one full turn about a datum axis. Total runout measures the combined effect of all cross-sections and the entire surface simultaneously. Total runout is a tighter, more comprehensive control used for precision rotating parts such as pump impellers and spindles.

#425 · Common Occupational Skills

What is surface roughness Ra and how is it measured?

Answer

Ra is the arithmetic average roughness, the average height of surface peaks and valleys over a measured length. It is measured with a surface profilometer (contact stylus instrument) in micrometres or microinches. A lower Ra number means a smoother surface. Bearing seats and seal contact surfaces require low Ra values to prevent leaks and premature wear.

#426 · Common Occupational Skills

What is the difference between bilateral and unilateral tolerances on a drawing?

Answer

A bilateral tolerance allows variation in both directions from nominal, for example 2.000 plus or minus 0.005 (meaning the part can be anywhere from 1.995 to 2.005). A unilateral tolerance allows variation in one direction only, such as 2.000 plus 0.000 minus 0.010. Unilateral tolerances are common on fits where one side of the range is critical.

#427 · Common Occupational Skills

What is the hole basis fit system and why is it preferred in manufacturing?

Answer

In the hole basis system, the hole is kept at nominal size (the lower deviation of the hole is zero) and the shaft size is varied to achieve the required fit. It is preferred because standard drill and reamer sizes produce consistent holes, making it easier and cheaper to hold the hole at a fixed size than to hold the shaft.

#428 · Common Occupational Skills

What does an ISO fit designation of H7/p6 mean on a drawing?

Answer

H7 is the hole tolerance: H means hole basis (lower deviation is zero) and 7 is the tolerance grade (IT7). p6 is the shaft tolerance: p means the shaft is oversized relative to the hole (an interference fit) and 6 is the tolerance grade (IT6). H7/p6 is a press fit used for bearing outer rings pressed into housings.

#429 · Common Occupational Skills

What is the difference between a clearance fit and an interference fit?

Answer

In a clearance fit the shaft is always smaller than the hole, leaving a gap that allows relative motion (used for running fits, sliding fits, and locating fits). In an interference fit the shaft is always larger than the hole, so assembly requires pressing or heating, and the parts hold together by friction (used for press-fit hubs, bushings, and bearing races).

#430 · Common Occupational Skills

How do you read a 0-to-1-inch outside micrometer to the ten-thousandths?

Answer

Read the sleeve scale (each numbered line is 0.100 inch, each unnumbered line is 0.025 inch), then add the thimble scale (each graduation is 0.001 inch), then add the vernier scale on the sleeve (each graduation is 0.0001 inch). Add all three numbers together. Practicing on a set of standards until readings are repeatable is essential for journeyperson-level work.

#431 · Common Occupational Skills

How do you read a vernier caliper to 0.001 inch?

Answer

Read the main scale to the last full 0.025-inch increment before the vernier zero line, then find which vernier graduation aligns exactly with a main scale line and multiply that number by 0.001 inch. Add both numbers together. Zero the caliper on a gauge block before critical measurements to confirm it reads correctly.

#432 · Common Occupational Skills

How is a dial bore gauge (telescoping gauge and mic) used to measure a bore?

Answer

A telescoping gauge is compressed, inserted into the bore at the tightest point, locked, and then withdrawn. The locked dimension is then measured across the gauge pins with an outside micrometer. This two-step process transfers the bore size to the micrometer. Multiple measurements at different depths and angles check taper and out-of-round.

#433 · Common Occupational Skills

What is a dial indicator used for in millwright alignment and assembly work?

Answer

A dial indicator measures small linear displacements, typically in 0.001-inch increments, by amplifying the movement of a contact point through gears to a needle on a face. Millwrights use it to check shaft runout, coupling face runout, bearing end-play, and to verify alignment corrections during shaft alignment procedures.

#434 · Common Occupational Skills

What is a surface plate used for and how is its accuracy maintained?

Answer

A granite or cast-iron surface plate provides a reference flat surface for layout, inspection, and precision measurement. Parts are placed on it to check flatness, set up V-blocks, or use height gauges. Accuracy is maintained by keeping it clean, never placing heavy or rough items on it without a soft pad, and having it certified flat on a regular calibration schedule.

#435 · Common Occupational Skills

What is a V-block used for when measuring or inspecting round shafts?

Answer

A V-block holds a round shaft or bar securely at two contact points so it can be rotated for runout checks with a dial indicator, or held still for layout work. The V angle (usually 90 degrees) centres the round part automatically regardless of diameter within the block's range. V-blocks always come in matched pairs.

#436 · Common Occupational Skills

How is an angle plate used in layout and machining setup work?

Answer

An angle plate is a precision-ground cast-iron or steel block with two faces at exactly 90 degrees to each other. It is clamped to a surface plate or machine table to hold a workpiece vertically for layout, drilling, or milling. Slots in the angle plate allow clamps to secure parts of various sizes.

#437 · Common Occupational Skills

What physical hazard classes does WHMIS 2015 recognize?

Answer

WHMIS 2015 adopted the GHS framework and recognizes 16 physical hazard classes, including: flammable gases, flammable aerosols, oxidizing gases, gases under pressure, flammable liquids, flammable solids, self-reactive substances, pyrophoric liquids and solids, self-heating substances, substances that emit flammable gas on contact with water, oxidizing liquids and solids, organic peroxides, corrosive to metals, desensitized explosives, combustible dusts, and simple asphyxiants. Each class has a distinct GHS pictogram.

#438 · Common Occupational Skills

What health hazard classes does WHMIS 2015 recognize?

Answer

WHMIS 2015 adopts the GHS framework and recognizes 10 health hazard classes: acute toxicity, skin corrosion and irritation, serious eye damage and eye irritation, respiratory and skin sensitization, germ cell mutagenicity, carcinogenicity, reproductive toxicity, specific target organ toxicity (single exposure), specific target organ toxicity (repeated exposure), and aspiration hazard. Each class has sub-categories rated from Category 1 (most severe) downward.

#439 · Common Occupational Skills

What are the 16 sections of a Safety Data Sheet (SDS)?

Answer

The 16 sections are: identification; hazard identification; composition; first-aid measures; fire-fighting measures; accidental release; handling and storage; exposure controls and PPE; physical and chemical properties; stability and reactivity; toxicological information; ecological information; disposal; transport; regulatory; and other information. Sections 1 through 8 are the most critical for day-to-day safe handling.

#440 · Common Occupational Skills

What PPE is required for grinding operations in a millwright shop?

Answer

Grinding requires a face shield (full face protection against flying sparks and particles), safety glasses underneath the face shield, hearing protection (grinding creates high noise levels), leather gloves, and leather or fire-resistant clothing to protect against sparks. Steel-toed boots are always required in the shop. Remove loose clothing or jewelry before grinding.

#441 · Common Occupational Skills

When is a full-face respirator required instead of a half-face respirator?

Answer

A full-face respirator is required when the contaminant can irritate or damage the eyes as well as the lungs (such as hydrogen sulfide, chlorine, or spray mist), when concentrations exceed the protection factor of a half-face unit, or when eye protection cannot be adequately combined with a half-face mask. The SDS will specify the minimum respiratory protection required.

#442 · Common Occupational Skills

What is the difference between a permit-required and a non-permit confined space?

Answer

A permit-required confined space has one or more serious hazards: hazardous atmosphere, engulfment risk, converging walls that could trap a person, or any other recognized serious safety hazard. Entry requires a written permit, atmospheric testing, a trained attendant, and a rescue plan. A non-permit space has no such hazards and does not require a formal entry permit.

#443 · Common Occupational Skills

What is the correct atmospheric testing order before confined space entry?

Answer

Test in this order: first oxygen level (must be 19.5 to 23.5 percent), then combustible/flammable gases (must be below 10 percent of lower explosive limit), then toxic gases such as hydrogen sulfide or carbon monoxide. Testing oxygen first ensures the combustible gas sensor reads accurately, since sensors can fail in low-oxygen environments.

#444 · Common Occupational Skills

List the six steps of lockout/tagout (LOTO) in order.

Answer

Step 1: identify all energy sources. Step 2: notify affected employees. Step 3: shut down the equipment using normal stopping procedure. Step 4: isolate all energy sources (close valves, open disconnects, block stored energy). Step 5: apply personal locks and tags. Step 6: verify zero energy state by attempting to start and checking pressure gauges and bleed points.

#445 · Common Occupational Skills

What does LOTO verification mean and why must it be done every time?

Answer

Verification means proving that all energy has been removed and cannot return before you begin work. This includes pressing the start button (and confirming nothing moves), checking that pressure gauges read zero, and confirming that bleed valves remain open. Verification must be done every time because the absence of a lock is not proof that energy is absent.

#446 · Common Occupational Skills

What are the key elements of a hot work permit?

Answer

A hot work permit identifies the location and type of work (welding, cutting, grinding), lists the combustible materials within the area and the measures taken to protect them (removal, wetting, shielding), requires atmospheric testing results, names the fire watch person and their duties, and states the permit start and expiry time. The fire watch must remain for at least 30 minutes after work ends.

#447 · Common Occupational Skills

What is the minimum anchor point rating for a personal fall arrest system?

Answer

A personal fall arrest system anchor point must be rated for a minimum of 22.2 kilonewtons (5,000 pounds) per person attached, or be certified by a professional engineer as being able to withstand the maximum arrest force. The anchor must be located above the worker's dorsal D-ring to minimize free fall distance.

#448 · Common Occupational Skills

How do you calculate total fall distance to select a lanyard for fall protection?

Answer

Total fall distance equals free fall distance (distance fallen before the lanyard engages, maximum 1.8 metres for a standard lanyard) plus deceleration distance (typically 1.2 metres for a shock-absorbing lanyard) plus the worker's height (approximately 1.8 metres from feet to D-ring). Add a safety margin of 0.6 metres. The anchor must provide enough clearance above the landing level.

#449 · Common Occupational Skills

What is the area formula for a triangle and when is it used in the trade?

Answer

Area of a triangle equals one-half times base times height ($A = 0.5 \times b \times h$). Millwrights use this for calculating the cross-sectional area of a V-belt, a triangular gusset plate, or a tapered shim. It also appears when using the 3-4-5 rule for squaring layouts on large foundations.

#450 · Common Occupational Skills

What does the cosine ratio describe in a right triangle?

Answer

Cosine equals the length of the adjacent side divided by the hypotenuse (CAH). A millwright uses cosine to find the horizontal run of an inclined surface, the horizontal component of a sling load, or the effective bearing width of an angled contact. If the angle is 60 degrees, cosine is 0.500, meaning the horizontal side is half the hypotenuse.

#451 · Common Occupational Skills

How do you calculate the gear ratio of a two-gear mesh from tooth count?

Answer

Gear ratio equals the number of teeth on the driven gear divided by the number of teeth on the driver gear. A 20-tooth driver meshing with a 60-tooth driven gear gives a ratio of 3:1, meaning the driven gear turns once for every three turns of the driver. Speed and torque change in inverse proportion to the ratio.

#452 · Common Occupational Skills

How do you find the output RPM of a belt drive knowing pulley diameters and input RPM?

Answer

Output RPM equals input RPM times the ratio of driver pulley diameter to driven pulley diameter. For example, a 6-inch driver turning at 1,800 RPM driving a 9-inch driven pulley gives 1,800 times $\frac{6}{9}$, which equals 1,200 RPM. Belt speed stays constant, so the smaller pulley always turns faster than the larger one.

#453 · Common Occupational Skills

What is the volume formula for a sphere and where might a millwright use it?

Answer

Volume of a sphere equals $\frac{4}{3}$ times pi times radius cubed ($V = \frac{4}{3} \times \pi \times r^3$). A millwright might use this to calculate the capacity of a spherical pressure vessel or tank, or to estimate the weight of a steel ball used in a ball valve. For a 6-inch diameter sphere the radius is 3 inches, giving a volume of about 113 cubic inches.

#454 · Common Occupational Skills

What is a detail view on a drawing and why is it used?

Answer

A detail view is an enlarged circle or box pulled from a main view to show a small or complex feature at a larger scale with more dimensions. It is labeled (usually with a letter) and the scale is stated separately. Millwrights look for detail views when a keyway, thread, groove, or tight-tolerance feature cannot be clearly dimensioned on the main view.

#455 · Common Occupational Skills

What does a transition fit mean in the context of shaft and bore assembly?

Answer

A transition fit sits between a clearance fit and an interference fit. Depending on where the actual machined dimensions fall within their tolerances, the assembled pair might have a small clearance or a small interference. Transition fits are used for locating fits where a part must be positioned accurately but still needs to be removable without a press.

#456 · Common Occupational Skills

What does perpendicularity control in GD&T; and how is it verified?

Answer

Perpendicularity controls how much a surface, axis, or line can deviate from being exactly 90 degrees to a datum. It is verified by placing the part against the datum surface on a surface plate and measuring the deviation from 90 degrees with a dial indicator swept along the controlled surface. Tight perpendicularity is critical for bearing housings and mating flanges.

#457 · Common Occupational Skills

What does parallelism control in GD&T; and when is it important for a millwright?

Answer

Parallelism controls how much a surface or axis can deviate from being exactly parallel to a datum. It is important for machine bed ways, linear guide rails, and shaft mounting pads where non-parallel surfaces cause binding, uneven loading, or misalignment. It is measured with a dial indicator swept across the surface while the part rests on the datum reference.

#458 · Common Occupational Skills

What is concentricity in GD&T; and how does it differ from runout?

Answer

Concentricity controls whether the median points of a feature (the centre points of cross-sections) all lie within a cylindrical tolerance zone centred on a datum axis. Runout controls the surface directly and is much easier to measure with a dial indicator. In practice, runout is almost always specified instead of concentricity because it is simpler to verify.

#459 · Common Occupational Skills

What is the angularity GD&T; symbol used to control?

Answer

Angularity controls a surface or axis that is at a specified angle other than 0 or 90 degrees to a datum. The tolerance zone is two parallel planes (or a cylinder) at the exact nominal angle to the datum. It is used for angled mounting faces, beveled surfaces, and inclined bores where the angle is critical for mating part alignment.

#460 · Common Occupational Skills

What does the position GD&T; symbol control on a pattern of holes?

Answer

The position symbol controls how far the centre of a hole (or other feature) can be from its theoretically exact location. The tolerance zone is a cylinder centred on the true position, and the hole axis must stay inside that cylinder. It is the most commonly used GD&T; control for bolt patterns, pin locations, and any feature that must line up with a mating part.

#461 · Rigging Hoisting and Moving

What is the difference between IWRC and fibre core wire rope and when is each used?

Answer

IWRC (independent wire rope core) has a small wire rope at its centre, giving it about 7.5 percent higher breaking strength and better resistance to crushing under heavy loads. Fibre core wire rope has a synthetic or natural fibre centre that acts as a lubricant reservoir, making it more flexible but less crush-resistant. Use IWRC for overhead lifting and running lines; fibre core is better for standing rigging where flexibility matters more than crush resistance.

#462 · Rigging Hoisting and Moving

What is the difference between regular lay and lang lay wire rope?

Answer

In regular lay wire rope, the wires in each strand are twisted opposite to the direction the strands are twisted around the core, which makes the rope resist rotation and is most common for lifting. In lang lay, both the wires and the strands are twisted in the same direction, giving better fatigue resistance and abrasion resistance but making the rope more prone to rotation under load.

#463 · Rigging Hoisting and Moving

What are the wire rope discard criteria based on broken wires per lay length?

Answer

Discard a wire rope if it has six or more randomly distributed broken wires in one rope lay length, or three or more broken wires in one strand in one lay length. A lay length is approximately 6.5 times the rope diameter. These criteria apply to running ropes; standing ropes may have different discard thresholds per the manufacturer and applicable standard.

#464 · Rigging Hoisting and Moving

What other wire rope conditions require immediate removal from service?

Answer

Remove a wire rope if it shows kinking, birdcaging (strands pushed apart), heat damage (discolouration, brittleness), corrosion pitting on the outer wires, a reduction in diameter of more than 3 percent from nominal, or core protrusion. Any one of these conditions makes the rope unsafe even if the broken-wire count is within limits.

#465 · Rigging Hoisting and Moving

What is the purpose of a swivel in a rigging assembly and when should it not be used?

Answer

A swivel allows the load to rotate freely without transmitting torque to the wire rope or chain, preventing twist buildup in the rigging. However, a swivel must never be used in a way that allows the load line to unscrew the swivel jaw under load. Also, do not use a plain swivel with lang lay wire rope, as it will unlay the rope and reduce strength.

#466 · Rigging Hoisting and Moving

How does the angle of an eyebolt reduce its rated load capacity?

Answer

An eyebolt is rated at full capacity only when the load is applied in line with the shank (straight pull). At 45 degrees from the shank axis, capacity drops to 50 percent of rated load. At 90 degrees (pulling sideways on the eye), capacity drops to 25 percent. Always use shoulder pattern eyebolts for angular loads, never plain pattern eyebolts.

#467 · Rigging Hoisting and Moving

What are the 11 standard crane hand signals a signal person must know?

Answer

The 11 standard ASME/CSA crane hand signals are: hoist, lower, use main hoist, use whip line, bridge travel (both directions), trolley travel (both directions), stop, emergency stop, dog everything, and move slowly. The signal person must have clear visual contact with the operator at all times, and the operator follows only the designated signal person's directions.

#468 · Rigging Hoisting and Moving

What are the three levels of overhead crane inspection and their frequency?

Answer

Pre-use inspection is performed by the operator at the start of each shift and checks hooks, chains or wire rope, limit switches, brakes, and controls. Periodic inspection is monthly to quarterly depending on service class and checks structural members, fasteners, and electrical components. Annual (frequent and periodic combined) is a full documented inspection covering all components by a qualified person.

#469 · Rigging Hoisting and Moving

What is the difference between proof load and working load limit (WLL) for rigging hardware?

Answer

Working load limit (WLL) is the maximum load a piece of hardware is designed to carry in normal service. Proof load is the test load applied during manufacturing to verify the hardware will not deform or fail; it is typically twice the WLL for most hardware. Proof load is not a usable working load; it is a one-time factory quality check.

#470 · Rigging Hoisting and Moving

How does sling angle affect the tension in each leg of a two-leg bridle sling?

Answer

As the sling angle from the horizontal decreases (legs become more spread), the tension in each leg increases dramatically for the same vertical load. At 60 degrees from horizontal each leg carries 1.15 times the load per leg. At 45 degrees it is 1.41 times. At 30 degrees it is 2.0 times. Never rig with a sling angle below 30 degrees from horizontal.

#471 · Rigging Hoisting and Moving

What is the de-rating factor for a wire rope sling in a choker hitch versus a straight pull?

Answer

A wire rope sling in a choker hitch has a rated capacity of approximately 75 percent (0.75 times) of its straight vertical hitch rating, because the bend over the load concentrates stress on the wire rope and fitting. A basket hitch with both legs vertical doubles the capacity to 2.0 times the straight hitch rating.

#472 · Rigging Hoisting and Moving

How does a basket hitch sling angle affect the working load limit?

Answer

A basket hitch with both legs at 90 degrees (straight down) gives a WLL of 2.0 times the single-leg vertical rating. As the legs spread and the sling angle decreases, the WLL drops. At 60 degrees from horizontal the efficiency factor is 1.73. At 45 degrees it is 1.41. At 30 degrees it falls to 1.0, equal to a single vertical leg.

#473 · Rigging Hoisting and Moving

What is a spreader bar (equalizer beam) used for in rigging and what problem does it solve?

Answer

A spreader bar (or equalizer beam) is a rigid bar or beam suspended from the crane hook with slings attached at each end to lift long or fragile loads. It prevents the slings from squeezing the load inward as they angle toward the hook, distributes the load evenly across multiple pick points, and allows vertical sling legs even when the pick points are far apart.

#474 · Rigging Hoisting and Moving

What is the difference between a tugger and a come-along for moving loads horizontally?

Answer

A come-along (hand-operated ratchet lever hoist) is a manual tool used for light pulling or positioning loads, typically up to 3 tonnes, using lever action and a hand chain. A tugger is a powered drum hoist (electric or pneumatic) mounted to a fixed structure or crane that provides much higher line pull for moving heavy loads horizontally. Both require proper rigging and anchor points rated for the load.

#475 · Rigging Hoisting and Moving

What is the difference between a chain hoist and a chain fall and what are their uses?

Answer

A chain hoist (lever hoist) is operated by moving a lever back and forth and is used for tensioning, positioning, and pulling loads in any direction. A chain fall (chain block) is operated by pulling the hand chain and is designed primarily for vertical lifting only. Both use mechanical advantage through a gear reduction to multiply force, and both are rated with a WLL that must not be exceeded.

#476 · Rigging Hoisting and Moving

What are the safe working rules for a bottle (hydraulic) jack when lifting machinery?

Answer

Always place the jack on a firm, level surface. Never exceed the jack's rated capacity. Use hardwood or steel blocking under the jack base and under the load to spread the load and prevent sinking. Keep the ram extension within the manufacturer's limits (never extend past the safety collar). Always secure the load with blocking before working under it, as a jack is not a safe hold.

#477 · Rigging Hoisting and Moving

How do you use a jacking beam to lift a load that has no single central jack point?

Answer

A jacking beam is a structural beam placed under the load at two or more support points. Jacks push up on the jacking beam, which distributes the lift to multiple load contact points simultaneously. This prevents point loading damage to equipment bases, allows a load to be raised evenly without tilting, and is essential when lifting equipment over a long span.

#478 · Rigging Hoisting and Moving

What is the correct procedure for skidding a heavy machine using steel rollers?

Answer

Place steel pipe rollers under the machine skid or base plate (typically three or more rollers perpendicular to the direction of travel). Push the load with a come-along or chain fall, feeding the last roller from the rear to the front as the machine moves forward. Keep rollers parallel and evenly spaced. The floor must be clean, flat, and able to carry the load without cracking.

#479 · Rigging Hoisting and Moving

What is the stability triangle of a forklift and why does it matter?

Answer

The forklift stability triangle is the area between the two front drive wheels and the single rear steer axle pivot point. The combined centre of gravity of the forklift and its load must stay inside this triangle for the forklift to remain stable. Overloading, raising a load too high, or turning sharply moves the centre of gravity outside the triangle, causing a tip-over.

#480 · Rigging Hoisting and Moving

When are personnel allowed to ride in the forks or on a load of a forklift?

Answer

Personnel are generally prohibited from riding on the forks or on a load unless a purpose-built, rated work platform is securely attached to the forks per the manufacturer's instructions and applicable safety regulations. Even with an approved platform, the forklift must travel at walking speed and the operator must not leave the controls. Never use forks directly as an improvised man-lift.

#481 · Rigging Hoisting and Moving

What is the minimum safe clearance required from overhead power lines during crane operations?

Answer

Under most Canadian and provincial regulations, a minimum clearance of 3 metres (approximately 10 feet) is required from overhead power lines up to 75 kV. For higher voltages the clearance increases. Always contact the utility owner before beginning any lift near power lines. The safest approach is to have the utility de-energize the line before the lift.

#482 · Rigging Hoisting and Moving

What is a lift plan and when is one required before a crane pick?

Answer

A lift plan is a written document that describes the load weight and dimensions, the crane or hoist to be used, rigging configuration and hardware with WLLs confirmed, the pick and set-down points, ground bearing capacity, swing radius exclusion zones, and the signal person and rigging crew assignments. A formal lift plan is required for critical lifts (near capacity, over personnel, tandem crane picks) and is best practice for all non-routine lifts.

#483 · Rigging Hoisting and Moving

What is a shackle and what are the two main types used in rigging?

Answer

A shackle is a U-shaped steel rigging connector with a removable pin used to attach slings, blocks, and hardware. The two main types are the anchor (omega) shackle, which handles in-line and angular loads well, and the chain (D) shackle, which has a narrower bow and is used where the load is applied in-line only. Screw-pin shackles must have the pin moused or safety-wired when used in a running block.

#484 · Rigging Hoisting and Moving

What is a turnbuckle used for in rigging and what is its load rating rule?

Answer

A turnbuckle is a threaded tensioning device used to adjust the length and tension of a wire rope, chain, or rod. It has a left-hand thread on one end and a right-hand thread on the other so turning the body shortens or lengthens the assembly. The WLL is stamped on the body. Never use a turnbuckle as a primary load-bearing sling; it is a tensioning and adjustment device.

#485 · Rigging Hoisting and Moving

What is a hoist ring (swivel hoist ring) and why is it preferred over a plain eyebolt?

Answer

A hoist ring is a threaded lifting point with a bail that pivots and swivels in all directions to stay aligned with the sling regardless of lift angle, maintaining full rated capacity at any angle. A plain eyebolt drops to 25 to 30 percent capacity at 90 degrees. Hoist rings are preferred for lifting machinery that requires angular picks or where the direction of pull is not straight up through the shank.

#486 · Rigging Hoisting and Moving

What is right-hand lay wire rope and how is it identified on site?

Answer

Right-hand lay wire rope has its strands twisted clockwise (to the right) when you look along the rope away from you, like a right-hand screw thread. It is the most common type used in industry. To identify it, hold the rope vertically: if the strands slope upward to the right, it is right-hand lay. Left-hand lay is used in special applications such as multi-part reeving to counteract torque.

#487 · Rigging Hoisting and Moving

How do you determine the centre of gravity of an asymmetric load before rigging?

Answer

For an asymmetric load, use the manufacturer's drawing if available since it will often mark the centre of gravity. If not, balance test: lift slightly off the ground with a single pick and observe which end drops. Shift the pick point toward the heavy end until the load hangs level. Mark the balance point and confirm all rigging hardware is rated for the full load at that configuration.

#488 · Rigging Hoisting and Moving

What is the purpose of a tag line during a crane lift and who controls it?

Answer

A tag line is a rope (never wire rope) attached to the load to control rotation and prevent the load from swinging into structures or personnel during travel. It is controlled by a designated ground worker who keeps tension on it while walking alongside the moving load. The tag line must never be wrapped around hands or body parts; hold it only in the hands with a clear release path.

#489 · Rigging Hoisting and Moving

What is a pick plan and how does it differ from a lift plan?

Answer

A pick plan is the portion of a lift plan that focuses specifically on the rigging configuration: how the slings attach to the load, the attachment points, sling angles, hardware selected, and the WLL confirmation for each component. A lift plan is broader and includes the crane selection, ground conditions, swing path, crew roles, and emergency procedures. The pick plan is a subset of the full lift plan.

#490 · Rigging Hoisting and Moving

What inspection must be performed on a chain hoist before every use?

Answer

Pre-use inspection includes checking the load chain for twisted, stretched, cracked, or worn links (discard if any link is worn more than 10 percent of original diameter). Check the hook for cracks, deformation, or a damaged safety latch. Confirm the brake engages and holds under a test lift. Check that the hand chain is intact and moves freely. Tag and remove from service any hoist that fails inspection.

#491 · Rigging Hoisting and Moving

How do you rig a choker hitch on a cylindrical load such as a shaft or pipe?

Answer

Pass one end of the sling around the load and through the other end (or through the choker fitting), then attach the free end to the hook. The choker must grip the load at least 30 degrees from the bottom (the self-tightening angle). Do not place the choke point at a sharp corner or edge without using a corner protector, as edges can cut the sling under load.

#492 · Rigging Hoisting and Moving

What is the maximum number of degrees a shackle pin can be loaded off the straight line of pull?

Answer

A shackle should be loaded in the plane of the bow (straight through the pin axis). Side loading a shackle pin, where the pin is loaded at an angle to its axis, dramatically reduces its capacity and can cause the bow to distort and the pin to walk out. Where angular loads are unavoidable, use an appropriately rated anchor shackle and stay within the manufacturer's angular de-rating guidance.

#493 · Rigging Hoisting and Moving

What does mousing a hook mean and when is it required?

Answer

Mousing a hook means securing the safety latch with wire or a tie strap so it cannot accidentally open under the sling. It is required when there is any risk that vibration, side loading, or contact with the load could pop the latch open and release the sling. Standard practice in most jurisdictions requires mousing on all lifting hooks unless a self-closing, self-locking safety latch is fitted.

#494 · Rigging Hoisting and Moving

What is the safe-working rule for loading the tip of a crane hook?

Answer

A crane hook must never be tip-loaded (loaded at the very tip of the hook) because this concentrates stress at the weakest point and can straighten the hook open. The load must seat at the saddle (the deepest inside bend of the hook). Always size your rigging hardware so the load seats correctly and does not slide toward the tip under load.

#495 · Rigging Hoisting and Moving

How do dollies differ from rollers when moving heavy machinery?

Answer

Rollers are simple steel pipes placed under a skid and moved one at a time as the load travels forward. Dollies are wheeled platforms (steel or polyurethane wheels) that stay attached to the machine for the entire move, allowing steering and direction changes that rollers cannot provide. Dollies are faster and give more control but require a flat, hard floor and sufficient clearance under the machine base.

#496 · Rigging Hoisting and Moving

What does the term safe working load (SWL) mean compared to WLL?

Answer

Safe working load (SWL) is an older term meaning the same thing as working load limit (WLL): the maximum load a piece of equipment is designed to handle in normal service. WLL is the current preferred term in most Canadian and international standards. Both values already include the manufacturer's design safety factor; the operator must never exceed either without written engineering approval.

#497 · Rigging Hoisting and Moving

What is a tandem lift and what additional precautions does it require?

Answer

A tandem lift uses two cranes simultaneously to lift a single load that is too large or heavy for one crane. It requires a lift director coordinating both operators, a written lift plan with load distribution calculations for each crane, communication protocol between operators (radio), and both cranes must be rated for their share of the load including all dynamic factors. Tandem lifts are classified as critical lifts in all jurisdictions.

#498 · Rigging Hoisting and Moving

What are the main components of a wire rope end termination using a swaged socket?

Answer

A swaged (mechanical) socket termination consists of a steel socket with a tapered cavity and a steel ferrule. The rope end is inserted, the wires are spread or the ferrule is positioned, and a hydraulic press swages (compresses) the metal around the rope to form a permanent, full-strength connection rated at 100 percent of the rope's minimum breaking force. It cannot be inspected internally once installed and must be discarded if any cracking or distortion is found on the socket.

#499 · Rigging Hoisting and Moving

What is the correct way to store wire rope slings after use to extend their service life?

Answer

Clean the sling to remove dirt, grit, and corrosive materials. Lightly oil the wire rope to prevent corrosion. Hang slings on a rack (not piled on the floor) so they drain and dry. Store in a clean, dry area away from chemicals, heat, and direct sunlight. Inspect before storage and tag any sling that needs repair so it is not accidentally returned to service.

#500 · Rigging Hoisting and Moving

What information must appear on a rigging inspection tag attached to a sling?

Answer

A rigging inspection tag must show the sling identification number, the rated WLL for each hitch type, the date of last inspection, the name or initials of the inspector, and the next inspection due date. Any sling without a current valid inspection tag must be removed from service until inspected and re-tagged. This documentation requirement applies to wire rope, chain, and synthetic web slings.